When Trouble Travels: Peer Effects of Criminal Exposure in Schools

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Motivation

- Teenage years mark the peak of the first entry into the criminal justice system
  - First offenses peak at age 15 in England (Cattan et al., 2023a,b)
  - Damage to human capital (Aizer and Doyle Jr., 2015; Eren and Mocan, 2021)
  - High social costs: £1bn per year from youth violence crimes (Irwin-Rogers et al., 2020)

- Criminal/problematic behaviors spread through peers
  - Jail (Bayer et al., 2009; Ouss and Philippe, 2025; Damm and Gorinas, 2020; Johnsen and Khoury, 2024), Neighbors (Freeman et al., 1996; Grogger, 2002; Dustmann and Landersø, 2021; Billings and Schnepel, 2022; Domínguez, 2026; Disney et al., 2026), Key players (Lee et al., 2021), Risky behaviors (Powell et al., 2005; Nakajima, 2007; Carrell et al., 2008; Fletcher, 2012; Eisenberg et al., 2014)

- School is the most important place for children's social interactions
  - Compulsory education induces repeated interactions w/ school peers
  - Transition from non-offenders to offenders

This Paper

- I study the peer effects of offending in secondary school
  - Newly linked admin data on education and crime in England
- Identification relies on the timing of offenders' school transfers
  - Treatment is being exposed to an offender's school transfer into peers' school-grade for the first time
- Sizable peer effects
  - Prob. of starting to commit a crime increases by 12.9% in 4 months
  - Number of crimes increases by 24.3%
  - Changes in school norms could be an important channel
  - Social cost due to the peer effects is up to £20m
- A centralized school assignment scheme could reduce crime by 36% of the amount induced by peer effects
  - Trade-off: an assignment that minimizes total crimes may increase the number of new offenders

Contribution

- Peer effects in criminal activities
  - Freeman et al. (1996); Grogger (2002); Kling et al. (2005); Bayer et al. (2009); Stevenson (2017); Billings et al. (2019); Sviatschi (2022); Dustmann and Landersø (2021); Lee et al. (2021); Billings and Schnepel (2022); Dustmann et al. (2023); Byström (2025); Domínguez (2026); Lavy and Yancu (2025); Disney et al. (2026)

  $\Rightarrow$ Causal spillover effects of crime in school
  + Analysis on initiation of criminal offending
  + Discussing potential mechanisms

- School system/human capital accumulation and youth crime
  - Crimes deteriorate human capital
    - Becker (1962); Aizer and Doyle Jr. (2015); Eren and Mocan (2021)
  - Schooling reduces crimes
    - Lochner and Moretti (2004); Machin et al. (2011); Cook and Kang (2016); Landersø et al. (2017); Bacher-Hicks et al. (2024); Bell et al. (2022); Villa (2024)

  $\Rightarrow$ Contacts with offenders in school could damage peers' human capital

- Exclusion policy, optimal school assignment
  - Abdulkadiroğlu et al. (2005); McAra and McVie (2010); Kamada and Kojima (2015); Bacher-Hicks et al. (2024); Balestra et al. (2022)

  $\Rightarrow$ Simulate the optimal school transfer w.r.t. reducing crime
  Trade-off between the number of offenders and crimes

Road map

1. Background
2. Data
3. Identification
4. Results

Background

English State Education System

- 94% of children are in state schools (Kynaston and Green, 2019)
  - Primary schools (age 5-11)
  - Secondary schools (age 11-16)
  - Grade retention is extremely rare

- School Transfers
  - Admitted in general if a slot is available
  - Application process takes 2-3 months
  - Teachers and children in a classroom do not know whether they receive a transfer student or not until the day of transfer
  - Happens at the beginning of each term (September, January, and April)

Youth Crime and Punishment

- In England, children are responsible for their own crimes from the age of 10
- If a child commits a crime and is caught by police, the child is either
  - given a caution by police (guilty)
  - convicted with penalties (fine, probation, community order, etc.)
- Most youth crimes end up with a caution (80% for the first offense) (Cattan et al., 2025)
  - Common crimes:
    - Common assault and battery
    - Theft from shops
    - Damage to an allotment
- Youth crime records appeared on a background check for employment (DBS)

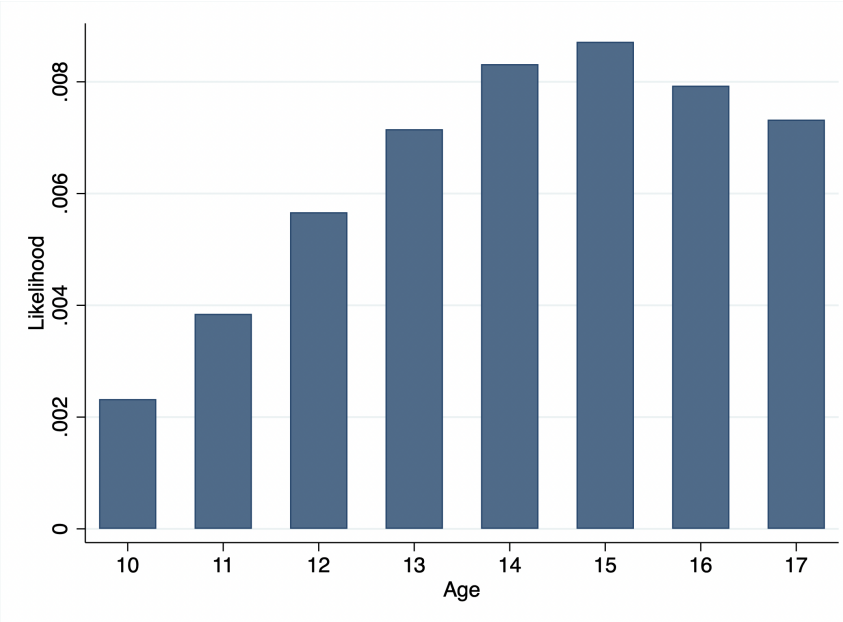
Data

Data

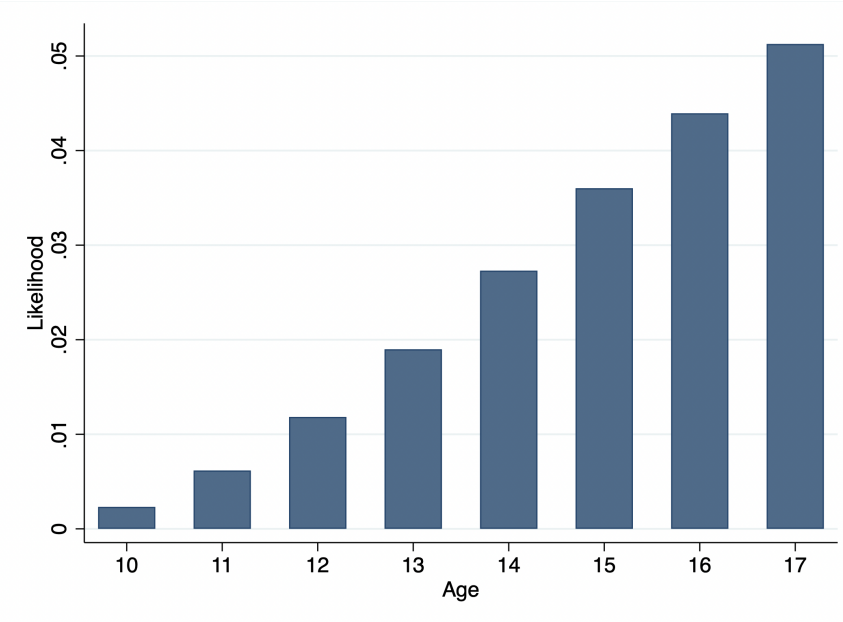
- National Pupil Database (NPD) for education and demographics
- Police National Computer (PNC) for crime records
- 1995/96 - 1998/99 Cohorts
  - 2.6 million obs in NPD
  - 2.1 million with control vars + in England for secondary school
  - 1.1 million who have ever been treated

▶ Sample Selection

First Offense Age



(a) Probability Mass Function



(a) Cumulative Mass Function

Timing of Offenders' Transfers

Age in Sept.	Offenders' Transfers		
	Autumn (1)	Spring (2)	Summer (3)
11	-	423	722
12	2,059	1,131	1,388
13	4,308	2,074	1,733
14	4,428	2,510	1,838
15	3,104	2,944	936

▸ Treatment Timing ▸ Non-offenders ▸ Offenders vs. Non-offenders
▸ Offenders' Characteristics

School Characteristics

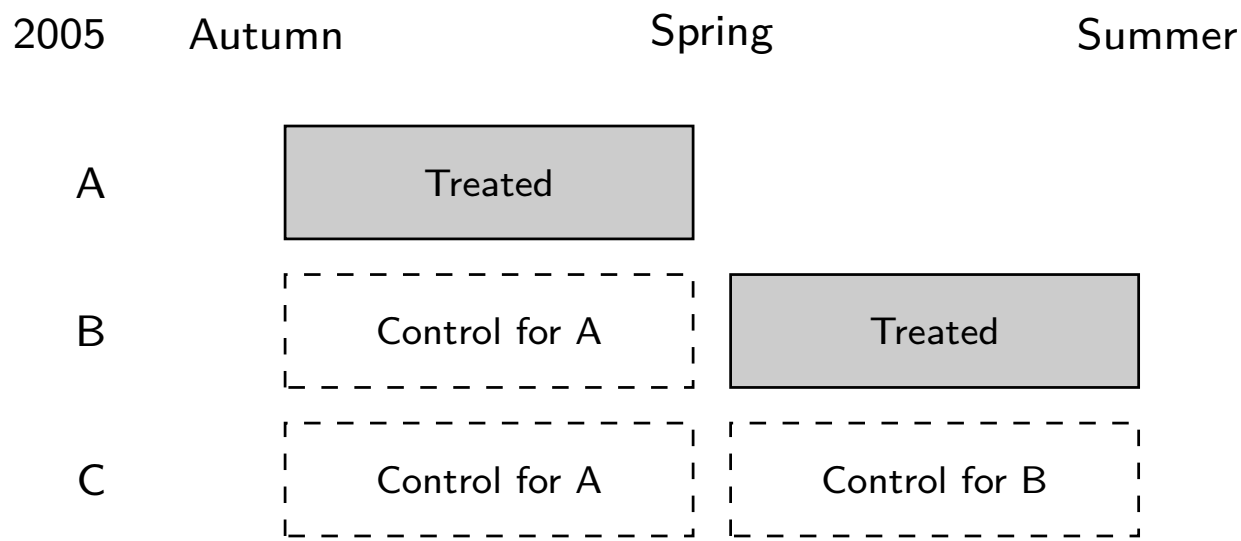
	Transfer (school-term) level		
	Offenders' transfers		
	Origin school (1)	Destination school (2)	Cond. Diff. (3)
Male	.536	.544	-.002**
	[.133]	[.122]	(.001)
Minority	.193	.185	-.002**
	[.235]	[.222]	(.001)
English as the first language	.890	.899	-.004***
	[.167]	[.153]	(.001)
Ever with SEN by age 11	.422	.444	-.014***
	[.143]	[.158]	(.001)
Ever eligible for FSM by age 11	.378	.397	-.006***
	[.188]	[.200]	(.001)
Age 7 score	-.132	-.168	.002
	[.307]	[.338]	(.002)
Age 11 score	-.201	-.227	.008***
	[.283]	[.283]	(.002)
Sample size	21,803	21,803	43,606

Notes: Conditional differences control for demographic characteristics: gender, minority status, first language status, SEN status, and FSM status. Leave-one-out estimations.

▸ All

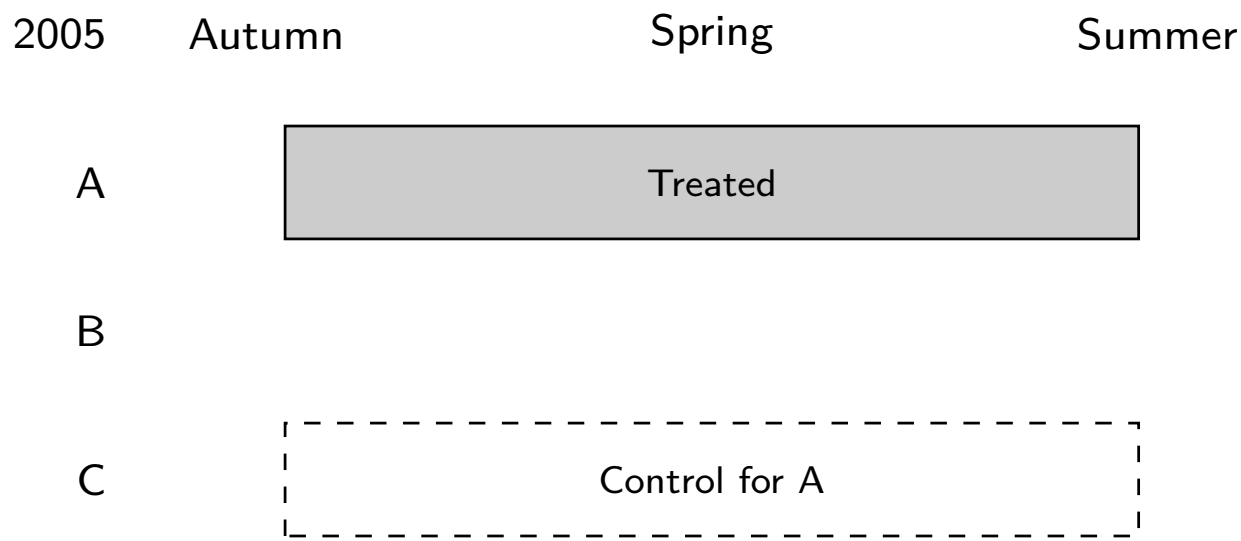
Treatment and Control Groups

- Treatment Group: children who are *already* exposed to an offender's school transfer into their school-grade
- Control Group: those who are *not yet* exposed to an offender's school transfer into their school-grade, but exposed by the end of secondary school



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Predicting Timing of Treatment

Supporting a valid comparison

Dummy for treatment by	Age 11 (1)	Age 12 (2)	Age 13 (3)	Age 14 (4)
Male	-.001*	.000	-.001	-.001***
	(.000)	(.001)	(.001)	(.000)
Minority	.000	.002	-.004*	.000
	(.001)	(.002)	(.002)	(.001)
English as the first language	.000	.000	-.001	-.001
	(.000)	(.001)	(.001)	(.000)
Ever with SEN by age 11	-.001	.003*	.000	.000
	(.001)	(.002)	(.002)	(.001)
Ever eligible for FSM by age 11	.000	-.001	-.001	.000
	(.000)	(.001)	(.001)	(.001)
F-test of joint test	.782	1.11	1.47	2.15
p-value	.584	.355	.185	.044
Sample size	1,050,567	1,050,567	1,050,567	1,050,567

Notes: I control for the linear month-of-birth term and the school and cohort fixed effects. The F-test also considers the coefficients of the linear month-of-birth term, which are insignificant at the 10% level for all regressions. Clustered at the treatment level.

▸ Predicting Treatment ▸ Early vs Late ▸ Even Split ▸ Age 11 vs 15 ▸ Snapshot

Identification

Identification: Staggered Event Study Design

- Exposure to others' criminal activities is endogenous

$\Rightarrow$ Timing of offenders' school transfers can be exogenous
  - Following Balsa et al. (2023) on effects of students w/ mental health problems.
  - Similar to Fadlon and Nielsen (2019, 2021)

- Staggered event-study framework (Callaway and Sant'Anna, 2021)
  - Controls include individual-level time-invariant characteristics and a linear age term
  - All standard errors are clustered at the treatment level (school x cohort x timing)

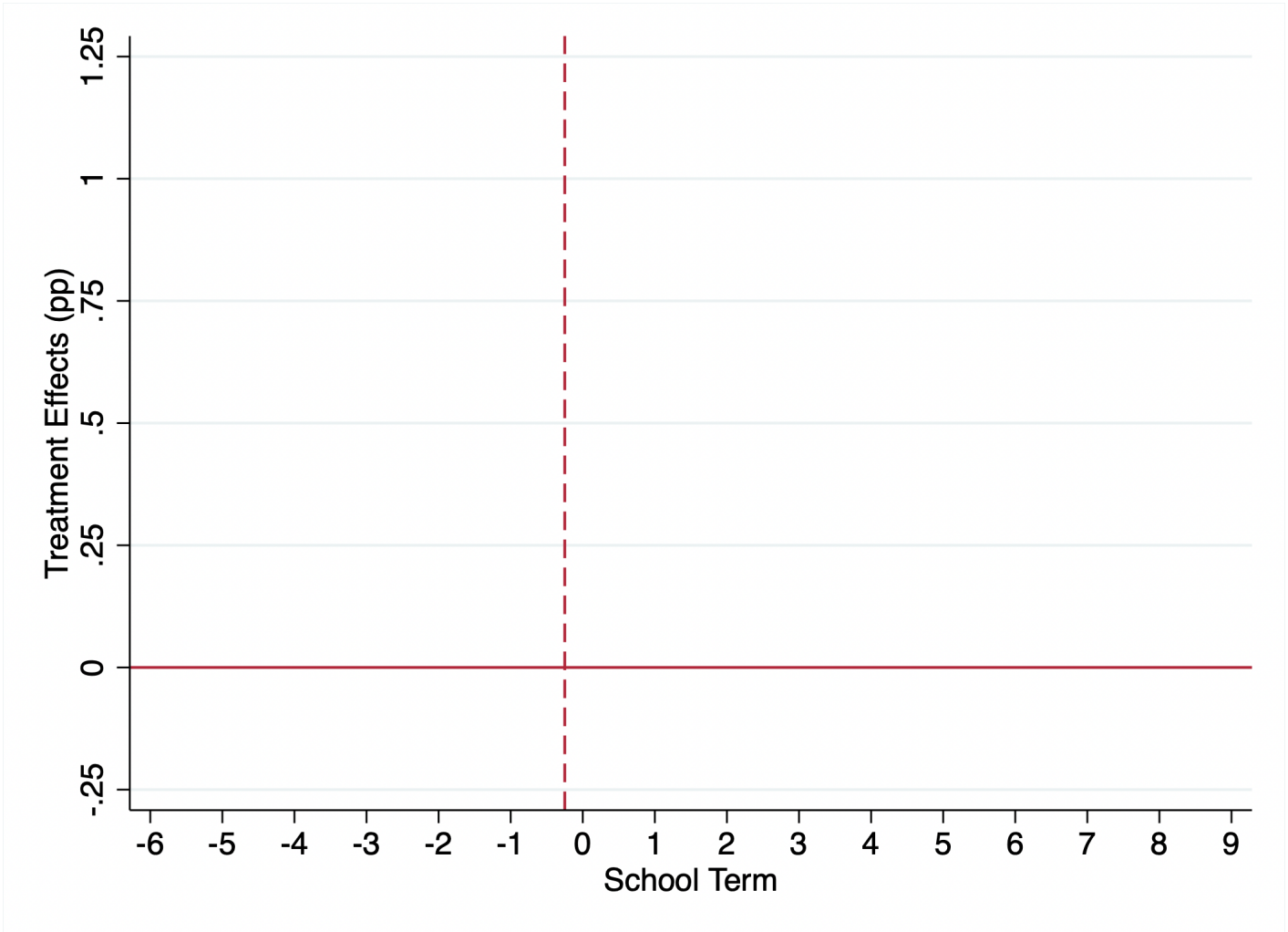
▸ Details

Identification: Key Assumptions

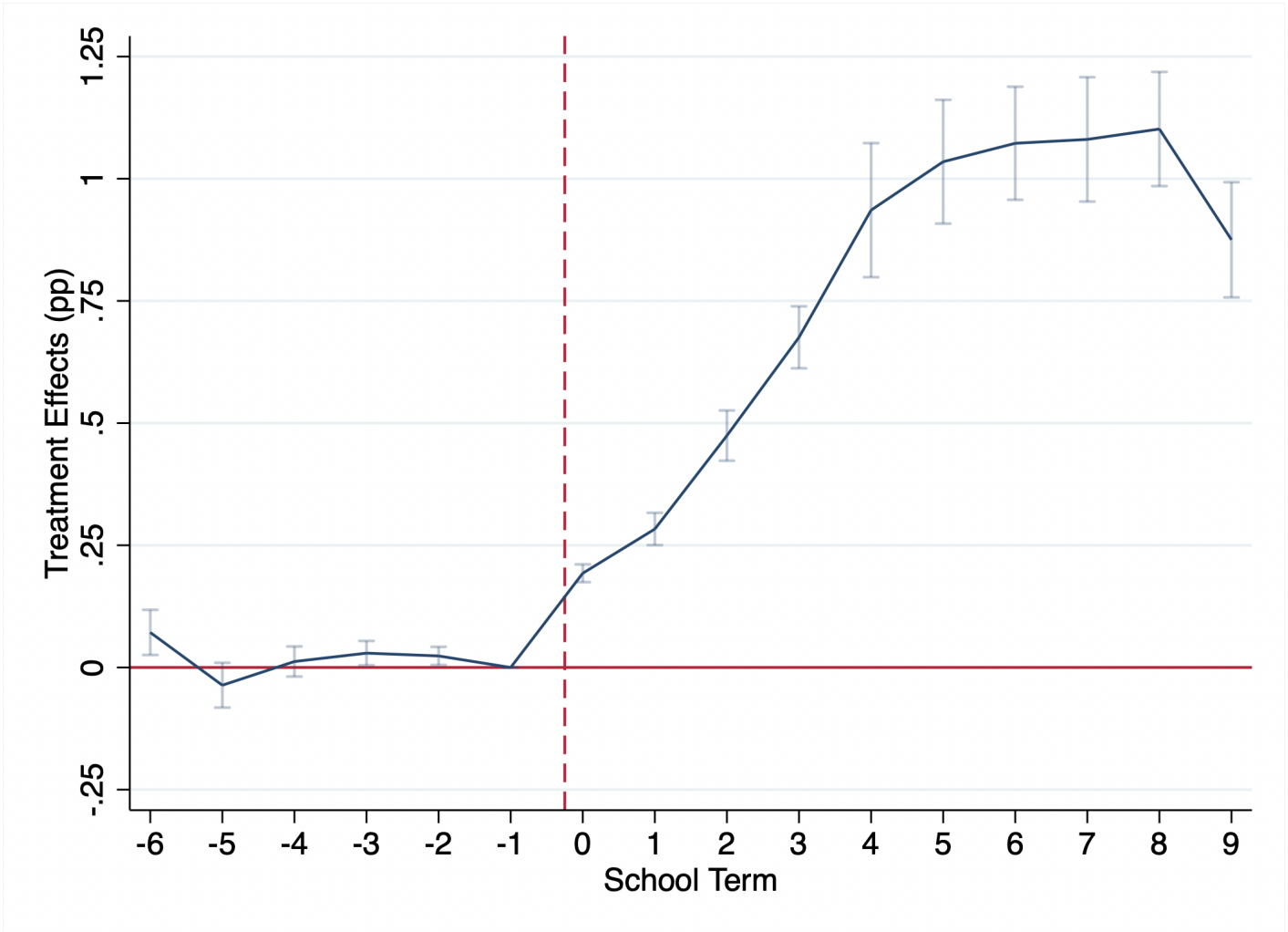
1. Conditional Parallel Trends
   - Without exposure to an offender's transfer, behavioral changes of children treated early and later would be the same
2. Limited Treatment Anticipation
   - Children who'll be exposed to the transfer later won't change their criminal behaviors prior to offenders' transfers.
   - Children, their parents, and teachers cannot predict offenders' school transfers.
3. Overlap
   - Each student has a positive probability of receiving an offender transfer, bounded by $(0, 1)$
   - I omit the never-treated group in the analysis.
4. Irreversibility of Treatment

Results

Peer Effects on Starting to Commit a Crime

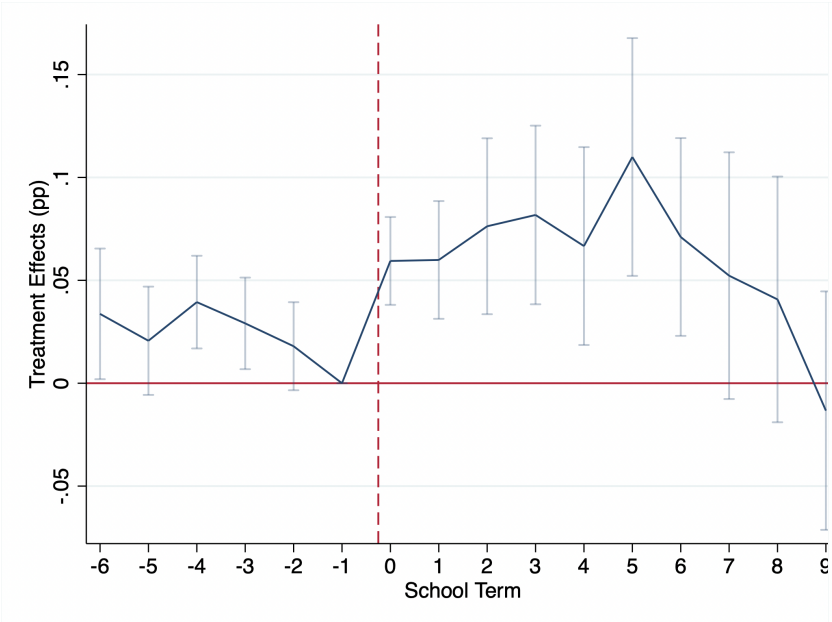


Peer Effects on Starting to Commit a Crime

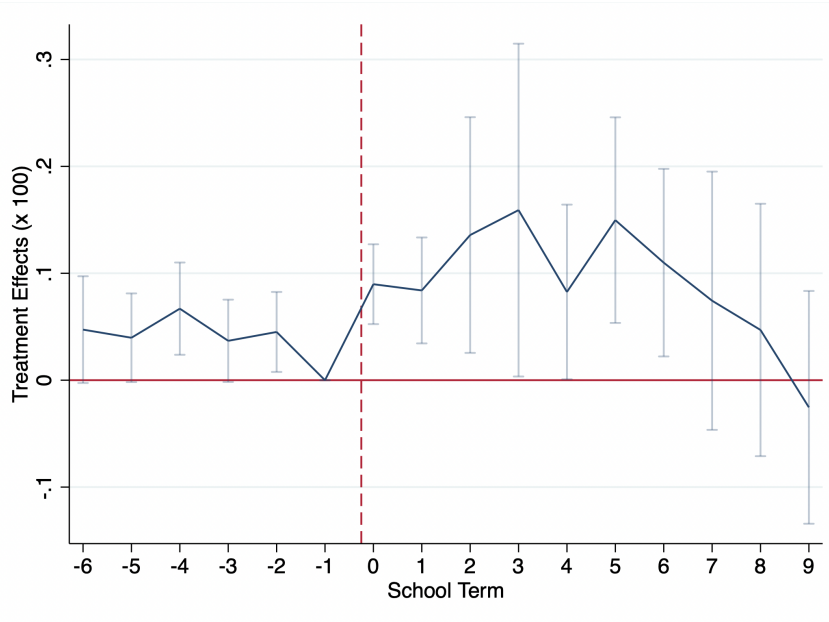


Peer Effects on Overall Criminal Activities

Including those who have started committing crime



(a) Any Crime at Each Period



(b) Number of Crimes at Each Period

Peer Effects of Crime

Effects on:	Crime Onset (1)	Number of Crimes (2)
Effects at $t = 0$ (x100)	.193***	.090***
	(.009)	(.019)
Pre-Treatment Mean (x100)	1.50	.369
Relative Effect Size	12.9%	24.3%
Panel Count	15,458,144	15,458,144
Individuals	1,050,567	1,050,567

Clustered at school-cohort-treatment timing level. *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Romano-Wolf step-down p-values are all smaller than 1%.

- (the effect size of crime onset) $\times$ (the number of students in the destination school) $= 0.00193 * 179 = 0.345$
- Every third offender's transfer adds one new offender

Robustness Checks

- Effects from not-yet-offenders
- Censored at peers' first school transfer
- Including never-treated group
- Long-distance transfers
- Effects on peers' school transfers
- Effects on those who stay in the school that the offender left
- Effects on misbehaviors

More

Treatment Intensity

A larger shock has a larger impact

- Intensity $= \dfrac{\text{(N of offenders transferred into a school-grade)}}{\text{(N of students in the school-grade)}}$

Intensity:	Lowest	Middle	Highest
Intensity (%)	.357	.510	.859
Treatment Effects at $t = 0$ (x100)	.099***	.123***	.325***
	(.014)	(.015)	(.022)
Pre-Treatment Mean (x100)	1.28	1.41	1.79
Relative Effect Size	7.68%	8.73%	18.1%
Sample Size (Panel Count)	5,165,416	5,027,003	5,265,725
Number of Individuals	350,482	333,455	348,441

School vs. Locality

Children interact with peers more in school

Locality Measure:	School	MSOA	LSOA
Treatment Effects at $t = 0$ (x100)	.193***	.070**	.050
	(.009)	(.030)	(.078)
Control Mean (x100)	1.50	1.25	1.46
Relative Effect Size	12.9%	5.60%	3.42%
School	Same	Different	Different
Locality	Any	Same	Same
Sample Size (Panel Count)	15,458,144	2,194,080	502,607
Number of Individuals	1,050,567	141,222	31,942

- MSOA: 5,000 to 15,000 residents
- LSOA: 1,000 to 3,000 residents

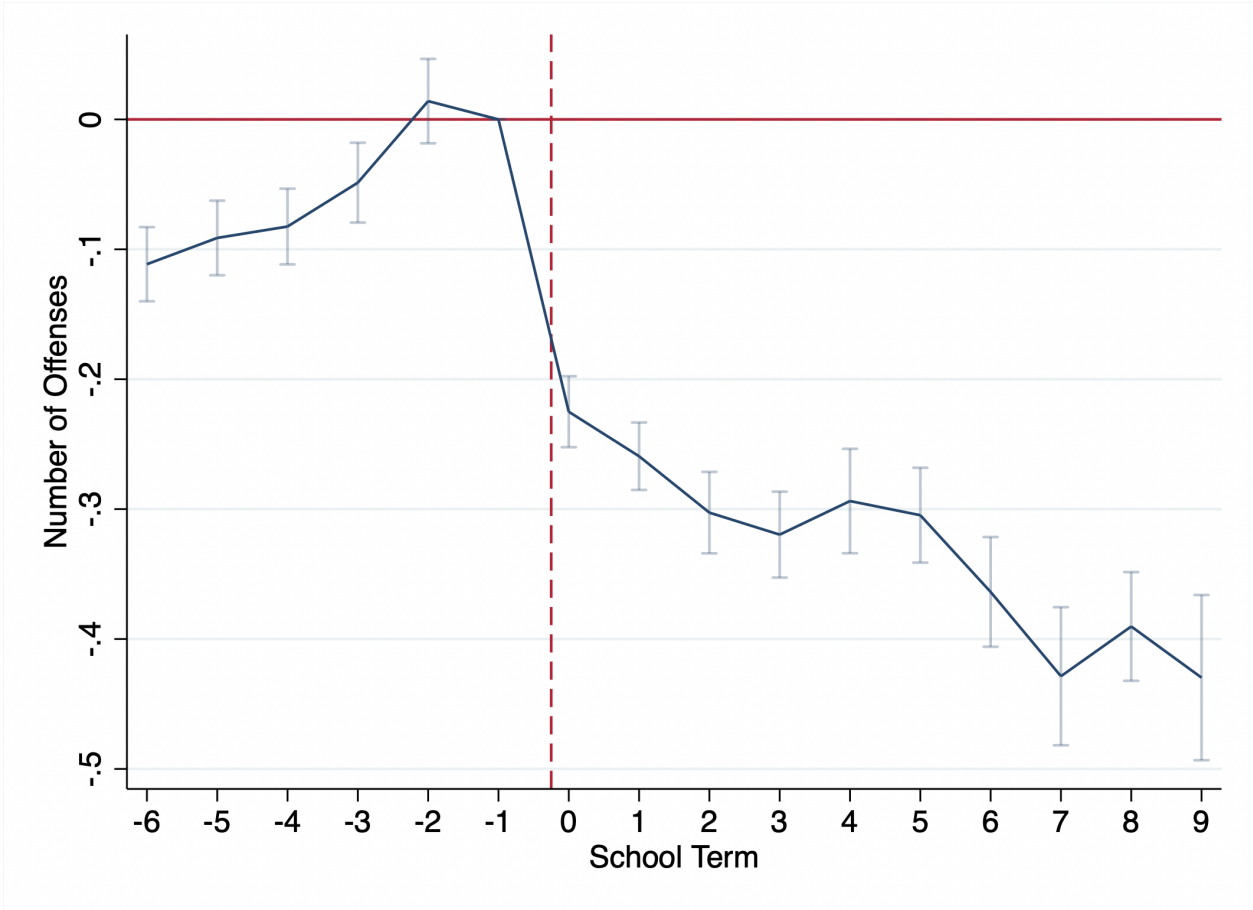
▸ Details ▸ School term vs Holiday

Potential Mechanisms

- Skill transmission
  - Peers commit the crime with transfer offenders
  - Peers commit the same type of crimes as the transfer offenders had committed
- Change in norms to tolerate offending
  - Peers might view the transfer offender as their role model
  - Peers are influenced by the altered social norm

Offenders' Behaviors Stop Offending After Transfer

Descriptive Evidence



Effects of Transfer on Number of Crimes

- Joint crime with transfer offenders does not explain the effects

▸ Joint Crime

Crime Type

Skill transmission is not a major channel

Crime Types:	All Crimes		Property Crimes	
	Same	Different	Same	Different
Treatment Effects at $t = 0$	.089***	.082***	.106***	.082***
	(.007)	(.008)	(.010)	(.009)
Pre-Treatment Mean	.714	.806	1.08	.714
Relative Effect Size	12.4%	10.2%	9.82%	11.5%
Sample Size (Panel Count)	12,184,487	12,184,487	8,351,676	8,351,676
Number of Individuals	827,507	827,507	567,629	567,629

Based on 4 crime categories: violence, property, drug-related, and other crimes.

▸ All types

Changes in School Norms could be a Key

- Change in norms in the destination schools might be an important channel
  - Offender-dense school
  - Seniority of transferring offenders
  - Schools with stricter discipline

Offender-dense School

A new offender in the least offender-dense school may change norms

- Density $= \dfrac{(\text{N of offenders at } t-1)}{(\text{N of students at } t-1)}$

Tercile in offender density:	Lowest	Middle	Highest
Panel A: Effects on Offending Initiation			
Treatment Effects at $t=0$	.023***	.243***	.034
	(.009)	(.055)	(.128)
Pre-Treatment Mean	.324	1.06	3.11
Relative Effect Size	7.15%	22.9%	1.11%
Panel B: Effects on the Number of Crimes			
Treatment Effects at $t=0$	-.012	.041	.284***
	(.015)	(.029)	(.097)
Pre-Treatment Mean	.100	.274	.738
Relative Effect Size	–12.0%	14.8%	38.5%
Sample Size (Panel Count)	5,101,909	5,084,637	5,049,704
Number of Individuals	344,858	344,858	344,858

Seniority

Senior peers are more likely to change the norm

Treatment Cohort:	Two Cohorts Younger	One Cohort Younger	Same Cohort	One Cohort Older	Two Cohorts Older
Treatment Effects at $t = 0$	.015	.051***	.193***	.204***	.073***
	(.011)	(.011)	(.009)	(.015)	(.026)
Pre-Treatment Mean	.400	.697	1.50	1.93	2.78
Relative Effect Size	3.65%	7.38%	12.9%	10.6%	2.61%
Sample Size (Panel Count)	6,704,483	9,790,276	15,458,144	4,737,409	1,398,616
Number of Individuals	453,765	663,904	1,050,567	322,733	95,296

Effects in Academy Schools

Academy schools have stricter discipline, so norm changes could be small

- An academy school is a state school with more autonomy
- Academy schools have tough discipline (Machin and Sandi, 2020)
- Converting schools to academies reduces crime (Duchini et al., 2020)
- Norm changes could be smaller in such schools

School Types:	Academy Schools (1)	Other Schools (2)
Treatment Effects at $t = 0$	.068**	.132***
	(.033)	(.010)
Pre-Treatment Mean	1.45	1.50
Relative Effect Size	4.69%	8.80%
Sample Size (Panel Count)	2,989,361	11,016,138
Number of Individuals	203,132	744,565

Standard errors reported in parentheses are clustered at the treatment level.

▸ Details

Potential Mechanisms

- Skill transmission is unlikely to be a major channel
- Together with Intensity results, change in norms to tolerate offending may be an important channel

Other Heterogeneity

Low opportunity costs amplify the treatment effects

- Early years non-cognitives
- Educational achievement
- Free School Meals and Special Education Needs
- Deprivation
- Demographics

Effects on Educational Achievement at Age 14

- Age 14 test score is available only for 1990/91-93/94 cohorts
- One school census a year

$$y_{ics} = \alpha + \beta d_{cs} + x_i'\delta + \eta_c + \nu_s + \varepsilon_{ics}$$

- $d = 1$ if exposed to an offender at the age of 13
- $d = 0$ if exposed to an offender at the age of 14

Transfer Student:	Offender	Non-offender
Treatment Effects	-.016***	.004
	(.005)	(.009)
Controls	x	x
Sample Size	862,553	69,297

Clustered at school-cohort level.

Statistical Mediation Analysis

Direct effect of exposure vs indirect effect through crime onset

$$y_i = \alpha + \beta d_i + \gamma c_i + \delta d_i * c_i + \varepsilon_i,$$

- y_i educational achievement of age 14
- d_i treatment dummy (studying in a disruptive environment)
- c_i compiler dummy (vulnerable to a shock of exposure to an offender)
- $d_i * c_i$ offender dummy at the timing of the exam

Coefficients:	γ	β	δ
Estimates	-.190***	-.016***	-.034***
	(.007)	(.002)	(.010)
Controls	x	x	x
Sample Size	397,031	808,936	827,222

Clustered at school-cohort level.

Social Cost

- Heeks et al. (2018) estimates the cost of each type of crime.

		Peer Effects		
N of Crimes	Total Cost	Δ N of Crimes	Δ Total Cost	Fraction
61,538	397,267,948	2,922	20,000,725	5.03%

Crime types used in this table are Violence with Injury, Violence without Injury, Rape, Other Sexual Offence, Robbery, Domestic Burglary, Theft of Vehicle, Theft from Vehicle, Theft from Person, Criminal Damage – Arson, Criminal Damage – Other, and Commercial Theft.

▸ Details

Simulation

Re-assigning offenders to a different school

- Given the heterogeneity in the treatment effects, there are ways to assign students to schools with less crime spillover
  - Under some constraints: availability and distance
  - Focus on the heterogeneity in offender-dense school results
  - Assume offenders and non-offenders are homogeneous
- Children without an offense history do not have an effect
- Assume students who want to move can move
- School that send offenders do *not* decrease crimes
- Use mechanism design theory

Recap: Offender-dense School

Clear trade-off between extensive and intensive margins

- Density $= \dfrac{(\text{N of offenders at } t-1)}{(\text{N of students at } t-1)}$

Tercile in offender density:	Lowest	Middle	Highest
Panel A: Effects on Offending Initiation			
Treatment Effects at $t=0$	.023***	.243***	.034
	(.009)	(.055)	(.128)
Panel B: Effects on the Number of Crimes			
Treatment Effects at $t=0$	-.012	.041	.284***
	(.015)	(.029)	(.097)
Sample Size (Panel Count)	5,101,909	5,084,637	5,049,704
Number of Individuals	344,858	344,858	344,858

Algorithm

1. Calculate the "potential" peer effects if a student transfers to each school with an available slot
   - Interpolate the effect size from offender-dense heterogeneity
   - No effects for non-offenders
   - Children cannot go to a far school
2. Stack them into a vector for each transfer student

Δ Crimes due to Peer Effects

	Offender A	Non-offender B
School 1	3	0
School 2	1	0

3. Re-interpret this vector as the student's preferences
   - Children want to go to school with the minimum spillover
   - If students cannot be allocated to another school, they will stay in the originally assigned school (outside option)
4. Apply standard matching methods like a random priority mechanism

(Bogomolnaia and Moulin, 2001)

Simulation results

Swapping offenders' and non-offenders' school transfers

		Swapping Radius (miles)					
	Status quo	1	3	5	10	25	Any
Panel A: Minimizing N of Crimes							
N of crimes	379	350	244	193	137	113	91
Change		-8%	-36%	-49%	-64%	-70%	-76%
N of offenders	887	893	917	932	924	918	826
Change		1%	3%	5%	4%	3%	-7%
Panel B: Minimizing N of Offenders							
N of crimes	379	392	498	609	805	1089	1244
Change		3%	31%	61%	113%	187%	228%
N of offenders	887	848	686	617	574	603	648
Change		-4%	-23%	-30%	-35%	-32%	-27%

▸ Details

Take Away and Broader Research Agenda

- I study the peer effects of offending in secondary school
- Identification relies on the timing of offenders' school transfers
- Sizable peer effects
  - Prob. of starting to commit a crime increases by 12.9% in 4 months
  - Social cost due to the peer effects is up to 5% of total youth crime
- A centralized school assignment scheme could reduce crime by 36% of the amount induced by peer effects

- Broader research agenda: education $\times$ crime $\times$ labor
  - Predicting prolific juvenile offenders (Cattan et al., 2025)
  - Gender differences in criminality (Cattan et al., 2023b)
  - Academies reduces crime (Duchini et al., 2020)
  - Early years, nursery, vocational training and offending
  - Effects of a criminal record on labor market outcomes

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- Broader research agenda: education $\times$ crime $\times$ labor
  - Prolific offenders and girls (Cattan et al., 2025, 2023b)
  - Educational and criminal trajectories throughout childhood to adolescence (Duchini et al., 2020)

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Appendix

Sample Selection

	Among NPD sample with controls Complete Records during Secondary School				Among Those in Column (1) Experienced a School-grade less than 20 Students				Among Those in Column (5) Ever Exposed to An Offender's School Transfer			
	Complete	Not Complete	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Never Experienced	Experienced	Uncond. Diff. [(5) = (6)]	Cond. Diff. [(5) = (6)]	Exposed	Never Exposed	Uncond. Diff. [(9) = (10)]	Cond. Diff. [(9) = (10)]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. Demographics												
Male	.510	.540	-.031***	-.019***	.506	.710	-.203***	-.124***	.518	.494	.024***	.004***
			(.001)	(.001)			(.002)	(.004)			(.001)	(.001)
Minority	.207	.500	-.293***	-.084***	.208	.191	.016***	.009***	.185	.231	-.046***	.000
			(.001)	(.001)			(.002)	(.003)			(.001)	(.000)
English as the first language	.892	.650	.242***	.058***	.892	.908	-.016***	-.018***	.902	.882	.019***	.001
			(.001)	(.001)			(.002)	(.002)			(.000)	(.000)
Ever with SEN by age 11	.351	.353	-.002**	.006***	.341	.941	-.600***	-.296***	.365	.316	.049***	.003***
			(.001)	(.001)			(.001)	(.004)			(.001)	(.001)
Ever eligible for FSM by age 11	.274	.322	-.049***	-.010***	.270	.508	-.238***	-.175***	.296	.243	.054***	.004***
			(.001)	(.001)			(.003)	(.005)			(.001)	(.001)
Offender by age 16	.214	.237	-.024***	-.025***	.210	.451	-.241***	-.555***	.243	.176	.067***	-.006***
			(.001)	(.001)			(.003)	(.004)			(.001)	(.001)
Any school transfer by age 16	1.17	1.26	-.091***	-.070***	1.16	1.30	-.144***	-.220***	1.18	1.14	.036***	-.012***
	[.439]	[.559]	(.002)	(.003)	[.431]	[.607]	(.005)	(.008)	[.445]	[.410]	(.001)	(.002)
N of school transfers by age 16 | any	.031	.135	-.104***	-.110***	.030	.124	-.094***	-.204***	.038	.021	.016***	.003***
			(.001)	(.001)			(.002)	(.004)			(.000)	(.000)
Age 5 score	.017	-.268	.285***	.080***	.029	–1.51	1.54***	.425***	-.011	.059	-.070***	.016
	[.992]	[1.08]	(.015)	(.013)	[.978]	[1.49]	(.057)	(.054)	[.967]	[.985]	(.007)	(.008)
Age 7 score	.025	-.357	.382***	.094***	.037	–1.84	1.88***	.503***	-.028	.105	-.133***	-.003
	[.990]	[1.08]	(.003)	(.002)	[.973]	[1.49]	(.013)	(.012)	[.959]	[.983]	(.001)	(.001)
Age 11 score	.026	-.387	.412***	.148***	.029	–1.14	1.17***	.418***	-.066	.140	-.206***	-.007***
	[.989]	[1.08]	(.004)	(.003)	[.987]	[1.15]	(.019)	(.019)	[.983]	[.981]	(.002)	(.002)
Age 16 score	.083	-.932	1.01***	.994***	.098	–2.00	2.09***	1.17***	.018	.178	-.160***	.001
	[.932]	[1.24]	(.003)	(.003)	[.916]	[.713]	(.006)	(.009)	[.912]	[.913]	(.001)	(.001)
Sample size*[1]	2,123,028	244,939	2,367,967	2,367,967	2,088,731	34,297	2,123,028	2,123,028	1,050,567	1,038,164	2,088,731	2,088,731

Back

Movers' Characteristics

	Non-movers (1)	Movers (2)	Cond. Diff. (3)
Male	.522	.507	.040***
			(.002)
Minority	.193	.160	-.013***
			(.001)
English as the first language	.897	.918	-.012***
			(.001)
Ever with SEN by age 11	.352	.409	-.061***
			(.002)
Ever eligible for FSM by age 11	.273	.368	-.131***
			(.002)
Offender by age 16	.031	.060	-.048***
			(.001)
N of school transfers by age 16 | any	-	1.18	-
		[.445]	
Age 5 overall score	.019	-.107	.032
	[.959]	[.985]	(.019)
Age 7 overall score	.001	-.126	.058***
	[.953]	[.976]	(.003)
Age 11 score (English, math, science)	-.016	-.225	.092***
	[.972]	[1.00]	(.004)
Age 16 score: point score | taken	.083	-.182	.320***
	[.887]	[.959]	(.003)
Sample size*[1]	795,011	255,556	1,050,567

Back

Sample Selection

	Ever Treated (1)	Never Treated (2)	Cond. Diff. (3)
Male	.518	.494	.004***
			(.001)
Minority	.185	.231	.000
			(.000)
English as the first language	.902	.882	.001
			(.000)
Ever with SEN by age 11	.365	.316	.003***
			(.001)
Ever eligible for FSM by age 11	.296	.243	.004***
			(.001)
Offender by age 16	.038	.021	.003***
			(.000)
Any school transfer by age 16	.243	.176	-.006***
			(.001)
N of school transfers by age 16 | any transfer	1.18	1.14	-.012***
	[.445]	[.410]	(.002)
Age 5 score	-.011	.059	.016
	[.967]	[.985]	(.008)
Age 7 score	-.030	.109	-.002
	[.961]	[.979]	(.001)
Age 11 score	-.066	.140	-.007***
	[.983]	[.981]	(.002)
Sample size*[1]	1,050,567	1,038,164	2,088,731

▸ All ▸ Movers ▸ Back

Offenders' Characteristics: All Offenses

	Offending Status by Age 16			
	Treaters	Non-treaters	Cond. Diff. [(1) = (2)]	Treaters, Before First Transfers
	(1)	(2)	(3)	(4)
Offense age	14.1	13.6	.551***	12.8
	[1.51]	[1.62]	(.008)	[1.46]
Caution	.526	.466	.064***	.690
			(.002)	
Theft/Robbery	.317	.317	-.010***	.336
			(.002)	
Battery	.169	.175	-.003	.190
			(.002)	
Drug Related	.064	.045	.015***	.029
			(.001)	
Violent	.131	.132	.003	.129
			(.002)	
Fraud	.022	.020	.002***	.017
			(.001)	
Weapons	.030	.027	-.000	.031
			(.001)	
Arson/Damage	.151	.166	-.009***	.191
			(.002)	
Sexual	.019	.016	.004***	.021
			(.001)	
Summary Motoring	.024	.022	.002***	.015
			(.001)	
Others/Unknown	.073	.080	-.004***	.042
			(.001)	
Sample size*[1]	155,327	62,143	217,470	35,109

Back

Timing of Treatment

Age in Sept.	Treatment		
	Autumn (1)	Spring (2)	Summer (3)
11	-	31,518	59,783
12	98,326	75,242	102,565
13	122,645	106,842	112,271
14	115,099	70,539	77,019
15	55,483	14,881	8,354

Back

Timing of Non-offenders' Transfers

Age in Sept.	Non-offenders' Transfers		
	Autumn (1)	Spring (2)	Summer (3)
A. Transfer			
11	-	18,856	30,998
12	87,990	22,859	31,001
13	146,986	24,126	19,517
14	88,152	20,340	14,235
15	40,334	16,364	10,217
B. First Contact to Others' Transfers (Treatment)			
11	-	1,269,477	434,327
12	270,010	44,740	19,738
13	20,164	8,026	2,775
14	3,798	1,704	682
15	2,255	10*1	

*1: There are 10 transfers for Spring and Summer at age 15 combined. Showing them separately is disclosive.

Offenders vs Non-offenders

	Offending Status by Age 16		
	Non-offenders (1)	Offenders (2)	Cond. Diff. (3)
Male	.511	.697	-.157***
			(.002)
Minority	.185	.171	-.007***
			(.001)
English as the first language	.901	.930	-.007***
			(.001)
Ever with SEN by age 11	.356	.604	-.148***
			(.002)
Ever eligible for FSM by age 11	.285	.591	-.235***
			(.002)
Any school transfer by age 16	.238	.387	-.091***
			(.002)
N of school transfers by age 16 | any transfer	1.17	1.30	-.093***
	[.434]	[.587]	(.005)
Age 7 overall score	.010	-.444	.117***
	[.927]	[.953]	(.004)
Age 11 overall score	-.042	-.577	.226***
	[.976]	[.971]	(.005)
Sample size	1,010,977	39,590	1,050,567

Back

Offenders' Characteristics

- Treaters: Offenders who moved to another school after their first crime

	Offending Status by Age 16		
	Treaters (1)	Non-treaters (2)	Cond. Diff. (3)
A. Demographics			
Male	.723	.706	.013***
			(.004)
Minority	.226	.185	-.003
			(.003)
English as the first language	.907	.936	-.007***
			(.002)
Ever with SEN by age 11	.629	.682	-.033***
			(.004)
Ever eligible for FSM by age 11	.607	.688	-.056***
			(.004)
B. Criminal Outcomes			
Age at first offense	13.8	12.5	1.30***
	[1.64]	[1.46]	(.012)
Number of offenses	2.65	3.66	-1.03***
	[3.76]	[4.99]	(.039)
Number of offenses per offending day	1.22	1.20	.012**
	[.658]	[.564]	(.005)
Sample size*[1]	72,859	20,953	93,812

▸ All Offenses ▸ Back

School Characteristics

		Transfer (school-term) level									
		Offenders' transfers				Non-offenders' transfers				Difference-in-differences	
	All school (1)	Origin school (2)	Destination school (3)	Uncond. Diff. [(2)=(3)] (4)	Cond. Diff. [(2)=(3)] (5)	Origin school (6)	Destination school (7)	Uncond. Diff. [(6)=(7)] (8)	Cond. Diff. [(6)=(7)] (9)	Uncond. Diff. [(4)=(8)] (10)	Cond. Diff. [(4)=(8)] (11)
Male	.531	.536	.544	-.008***	-.002**	.515	.516	-.000*	-.001***	-.008***	-.002
	[.174]	[.133]	[.122]	(.001)	(.001)	[.126]	[.125]	(.000)	(.000)	(.001)	(.001)
Minority	.216	.193	.185	.008***	-.002**	.192	.184	.008***	.000**	.000	-.003***
	[.256]	[.235]	[.222]	(.002)	(.001)	[.242]	[.231]	(.000)	(.000)	(.002)	(.001)
English as the first language	.885	.890	.899	-.009***	-.004***	.891	.896	-.005***	-.001***	-.004**	-.003***
	[.174]	[.167]	[.153]	(.002)	(.001)	[.174]	[.165]	(.000)	(.000)	(.002)	(.001)
Ever with SEN by age 11	.416	.422	.444	-.022***	-.014***	.387	.383	.003***	-.009***	-.026***	-.004***
	[.199]	[.143]	[.158]	(.001)	(.001)	[.122]	[.123]	(.000)	(.000)	(.001)	(.001)
Ever eligible for FSM by age 11	.331	.378	.397	-.019***	-.006***	.325	.319	.006***	-.005***	-.025***	.000
	[.197]	[.188]	[.200]	(.002)	(.001)	[.178]	[.173]	(.000)	(.000)	(.002)	(.001)
Early Years Score	-.152	-.013	-.015	.003	-.004	-.006	-.005	-.000	-.000	.003	-.002
	[.686]	[.060]	[.049]	(.003)	(.003)	[.046]	[.045]	(.000)	(.000)	(.003)	(.003)
KS1 Score (overall)	-.211	-.140	-.179	.039***	.003*	-.051	-.046	-.005***	.004***	.044***	-.006***
	[.737]	[.315]	[.348]	(.003)	(.002)	[.310]	[.320]	(.001)	(.000)	(.003)	(.002)
KS1 Score (English, Maths, Science)	-.210	-.132	-.168	.036***	.002	-.048	-.042	-.005***	.004***	.041***	-.007***
	[.744]	[.307]	[.338]	(.003)	(.002)	[.306]	[.316]	(.001)	(.000)	(.003)	(.002)
KS2 Score (English, Maths, Science)	-.180	-.201	-.227	.026***	.008***	-.138	-.127	-.010***	.001***	.036***	.003
	[.590]	[.283]	[.283]	(.003)	(.002)	[.280]	[.281]	(.001)	(.000)	(.003)	(.002)
Offender	.042	.054	.091	-.036***	-.028***	.022	.027	-.005***	-.006***	-.032***	-.027***
	[.101]	[.103]	[.150]	(.001)	(.001)	[.035]	[.047]	(.000)	(.000)	(.001)	(.001)
N Students	154	169	179	-9.28***	-16.2***	164	210	-46.3***	-45.4***	37.0***	31.1***
	[75.6]	[73.1]	[92.8]	(.800)	(.669)	[65.0]	[94.3]	(.153)	(.143)	(.815)	(.695)
Sample size*[1]	4,559	21,803	21,803	43,606	43,606	559,419	559,419	1,118,838	1,118,838	1,162,444	1,162,444

back

Treated and Control Group Balancing

	Treated at Age 11 (1)	Treated at Age 15 (2)	Cond. Diff. (3)
Male	.518	.505	.002
			(.004)
Minority	.153	.202	-.004
			(.003)
English as the first language	.920	.890	-.005**
			(.002)
Ever with SEN by age 11	.380	.353	-.016***
			(.004)
Ever eligible for FSM by age 11	.325	.274	-.005
			(.004)
Age 5 test score	-.027	.005	.137
	[.951]	[.970]	(.122)
Age 7 test score	-.068	.010	.002
	[.951]	[.970]	(.122)
Age 11 test score	-.136	.004	-.006
	[.989]	[.976]	(.014)
Sample size	91,301	78,718	170,019

Notes: Conditional differences control for demographic characteristics: gender, minority status, first language status, SEN status, and FSM status. Clustered at the treatment level.

Predicting Early Treatment

Male	-.001
	(.001)
Minority	-.004*
	(.002)
English as the first language	-.001
	(.001)
Ever with SEN by age 11	.000
	(.002)
Ever eligible for FSM by age 11	-.001
	(.001)
F-statistics of joint test	1.47
Sample size	1,050,567

Notes: Conditional differences control for demographic characteristics: gender, minority status, first language status, SEN status, and FSM status. Clustered at the treatment level.

Back

Balancing between the Treatment and Control Groups in Spring 2007

	Treated (1)	Control (2)	Cond. Diff. (3)
Male	.519	.518	-.002
			(.002)
Minority	.170	.198	.001
			(.001)
English as the first language	.910	.894	.000
			(.001)
Ever with SEN by age 11	.365	.364	-.003**
			(.001)
Ever eligible for FSM by age 11	.297	.293	.000
			(.001)
Age 5 test score	.002	.009	-.023
	[.948]	[.968]	(.028)
Age 7 test score	-.037	-.016	-.005
	[.962]	[.956]	(.003)
Age 11 test score	-.079	-.040	-.008**
	[.985]	[.978]	(.004)
Sample size	518,932	523,753	1,042,685

Notes: Control: cohort fixed effects and linear month-of-birth term, and secondary school FE. Clustered at the treatment level.

Back

Balancing between Early and Late Treated

Supporting a valid comparison

	Treated at Age 11-13 (1)	Treated at Age 14-15 (2)	Cond. Diff. (3)
Male	.519	.517	-.001
			(.001)
Minority	.178	.199	.000
			(.001)
English as the first language	.905	.894	-.001**
			(.001)
Ever with SEN by age 11	.372	.353	-.001
			(.001)
Ever eligible for FSM by age 11	.305	.278	-.001
			(.001)
Age 5 test score	-.019	.002	-.010
	[.969]	[.963]	(.020)
Age 7 test score	-.043	.003	-.002
	[.962]	[.953]	(.003)
Age 11 test score	-.085	-.023	-.003
	[.985]	[.976]	(.003)
Sample size	709,192	341,375	1,050,567

Notes: Conditional differences control for demographic characteristics: gender, minority status, first language status, SEN status, and FSM status. Clustered at the treatment level.

Back

Treated and Control Group Balancing

	Treated in Age 11-13 (1)	Treated in Age 13-15 (2)	Cond. Diff. (3)
Male	.520	.517	-.001
			(.001)
Minority	.174	.194	.001
			(.001)
English as the first language	.907	.897	.000
			(.001)
Ever with SEN by age 11	.376	.356	-.001
			(.001)
Ever eligible for FSM by age 11	.313	.282	-.002**
			(.001)
Age 5 test score	-.039	.009	-.023
	[.979]	[.957]	(.016)
Age 7 test score	-.059	-.002	-.003
	[.963]	[.956]	(.002)
Age 11 test score	-.102	-.033	-.007***
	[.987]	[.978]	(.003)
Sample size	490,079	560,488	1,050,567

Notes: Conditional differences control for demographic characteristics: gender, minority status, first language status, SEN status, and FSM status. Clustered at the treatment level.

Estimand

- The estimand is the average treatment effects on the treated (ATT):

$$ATT(g,t) = \mathbb{E}\left[\left(\frac{G_{ig}}{\mathbb{E}[G_{ig}]} - \frac{\frac{p_{g,t}(X_{ist})(1-D_{it})(1-G_{ig})}{1-p_{g,t}(X_{ist})}}{\mathbb{E}\left[\frac{p_{g,t}(X)(1-D_{it})(1-G_{ig})}{1-p_{g,t}(X_{ist})}\right]}\right) \times (y_{ist} - y_{isg-1} - m^{ny}_{g,t}(X_{ist}))\right],$$

G_{ig}: a dummy for whether individual i is first treated in period g

X_{ist}: Observed characteristics of individuals i, school s, at time t

D_{it}: Treatment status of individual i at time t

- population outcome regression for the "not-yet-treated":

$$m^{ny}_{g,t}(X_{ist}) = \mathbb{E}[y_{ist} - y_{isg-1} \mid X_{ist}, D_{it} = 0, G_{ig} = 0]$$

- Generalized propensity score:

$$p_{g,t}(X) = P(G_{ig} = 1 \mid X_{ist}, G_{ig} + (1 - D_{it})(1 - G_{ig}) = 1)$$

$$\theta(e) = \mathbb{1}\{g + e \leq \mathcal{T}\}P(G_{ig} = g \mid G_{ig} + e \leq \mathcal{T})ATT(g, g + e),$$

where $e = t - g$ denotes the time elapsed, and $\mathcal{T}$ is the last period in the data.

Problem of TWFE

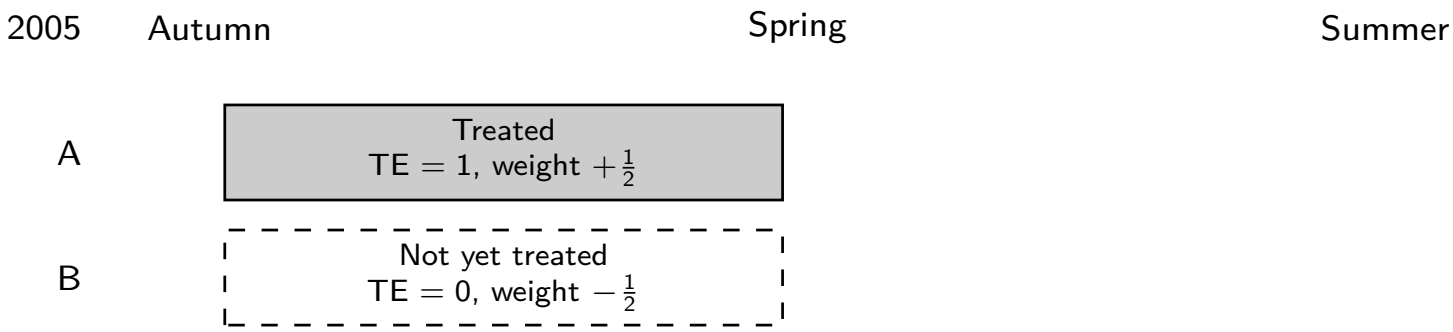
- Two groups: A and B
- Two periods: Autumn and Spring
- A is treated in Autumn, B is in Spring
- Assume no trend
- True treatment effects are all positive
  - A(Autumn) is 1
  - A(Spring) is 2
  - B(Spring) is 1
- TWFE regression:

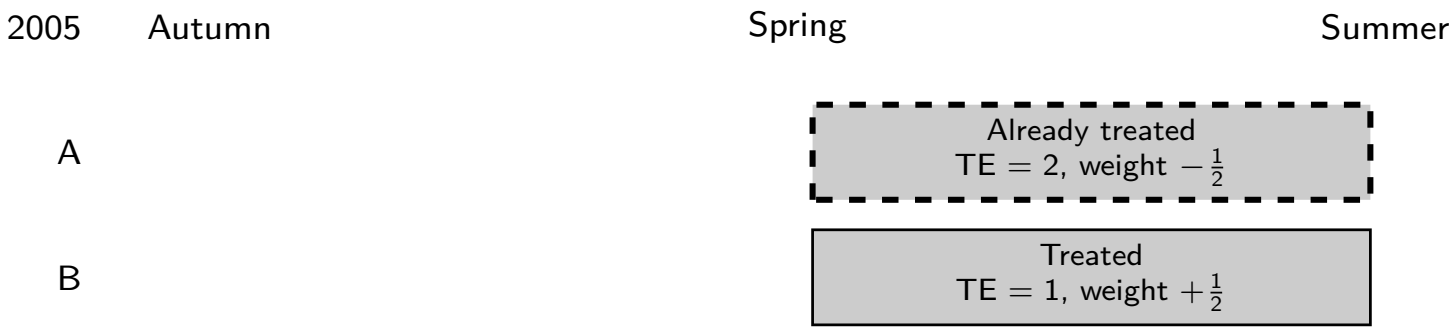
$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}$$

- $\beta^{TWFE} = 1 \times A(Autumn) + \left(-\frac{1}{2}\right) \times A(Spring) + \frac{1}{2} \times B(Spring) = 0.$

Back

Problem of TWFE

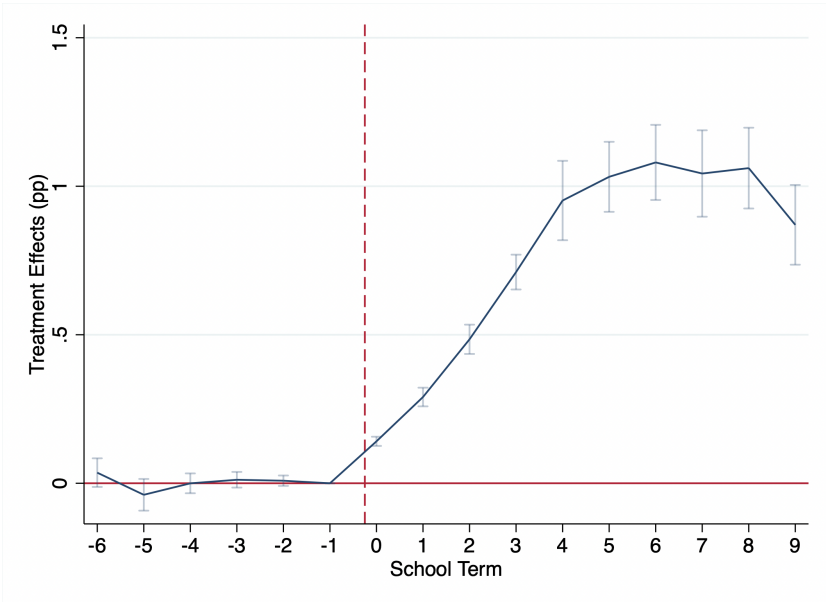




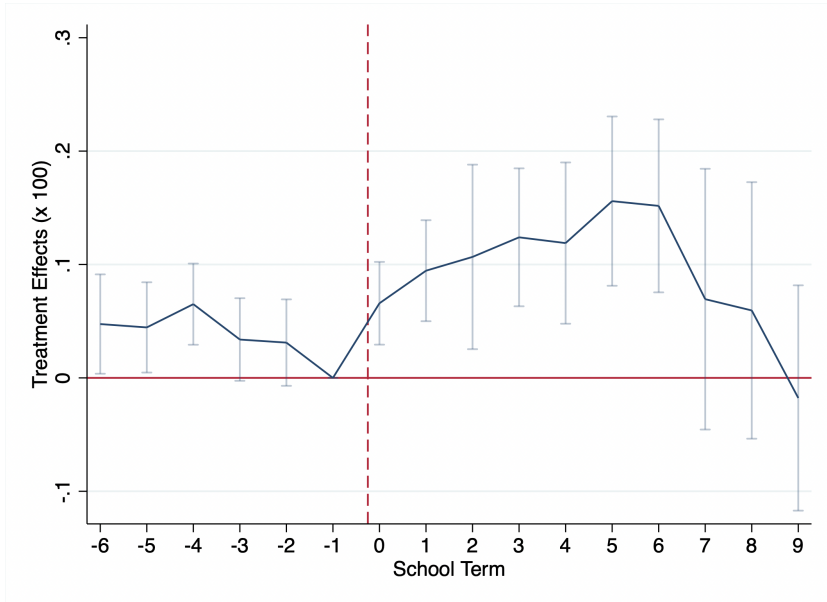
$$\hat{\beta}^{TWFE} = \frac{1}{2}\left(\Delta Y_A^{\text{Aut}} - \Delta Y_B^{\text{Aut}}\right) + \frac{1}{2}\left(\Delta Y_B^{\text{Spr}} - \Delta Y_A^{\text{Spr}}\right)$$
$$= \underbrace{\frac{1}{2}(1) - \frac{1}{2}(0)}_{\text{clean}} + \underbrace{\frac{1}{2}(1) - \frac{1}{2}(2)}_{\text{forbidden}} = 0$$

Back

Censored Records at the First Transfer



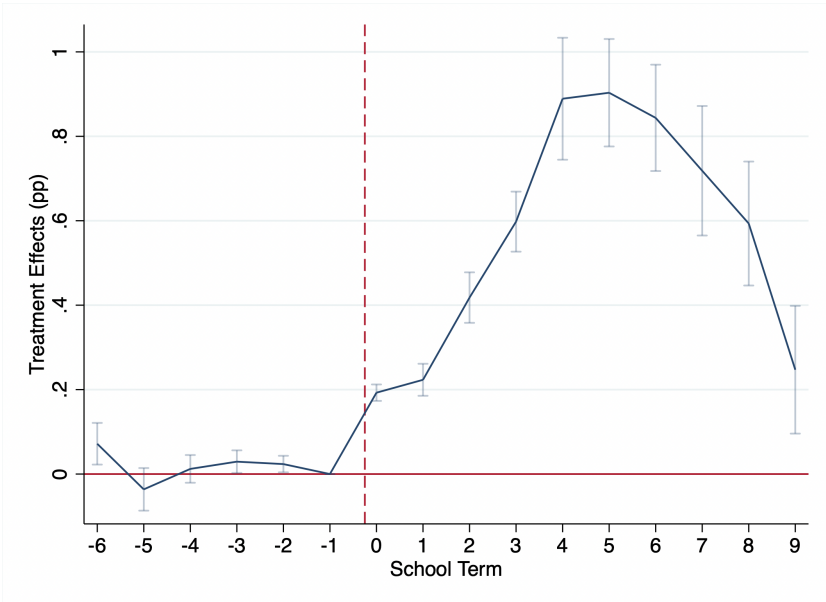
(a) Effects on Offending Initiation



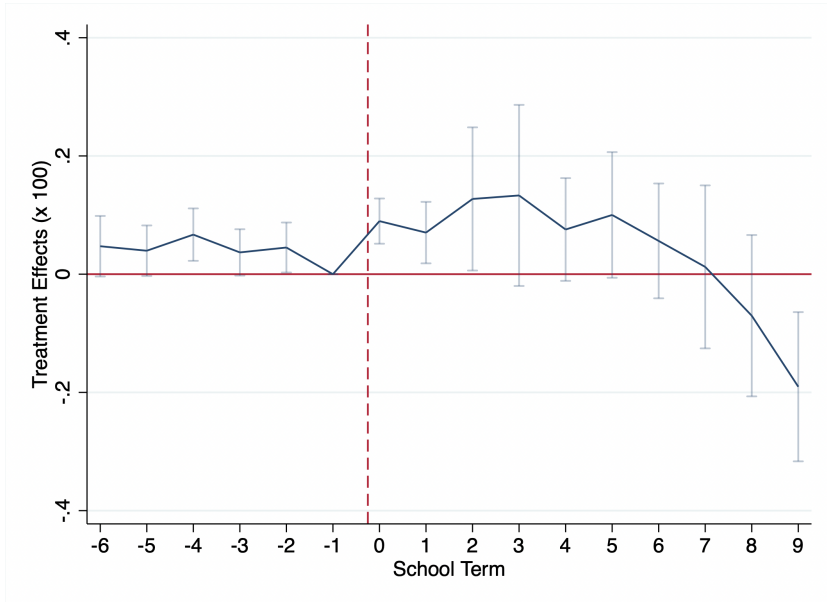
(b) Effects on Number of Crimes

back

Censored Records at the Second Treatment



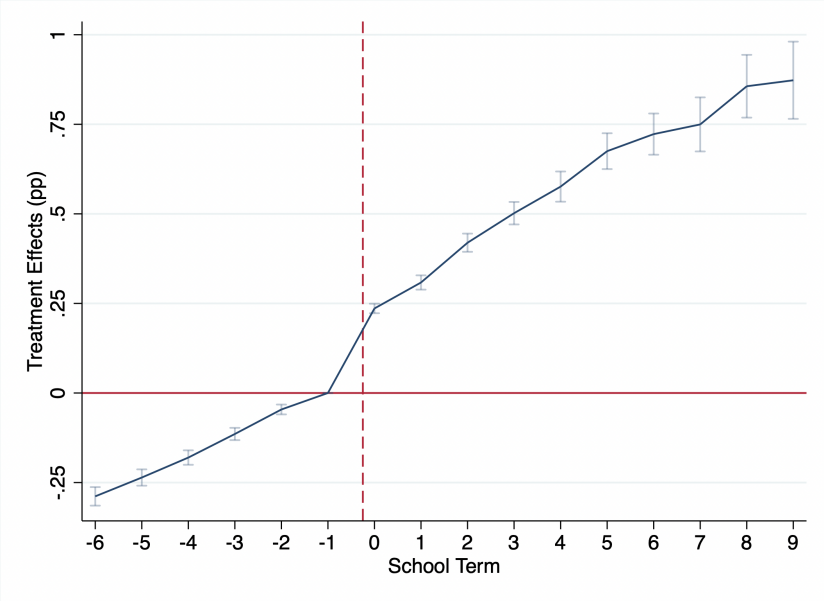
(a) Effects on Offending Initiation



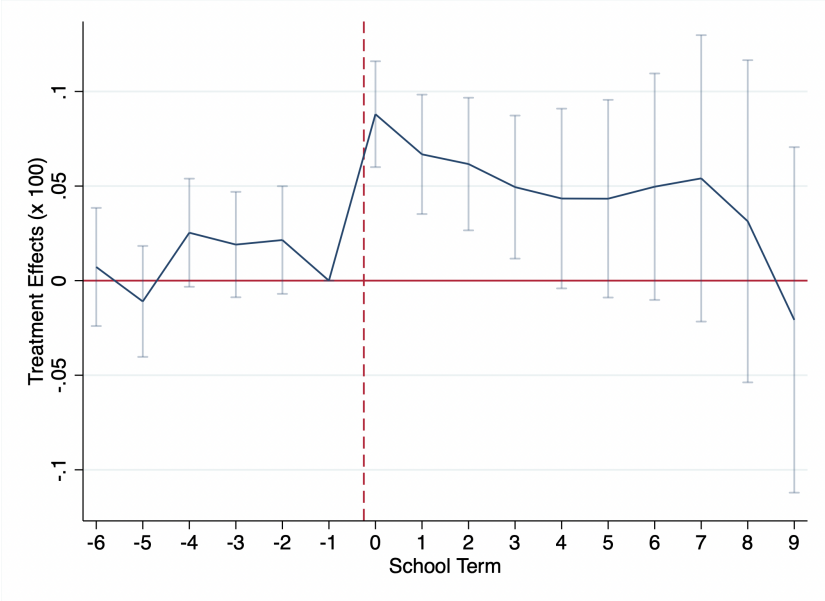
(b) Effects on Number of Crimes

back

Estimation without Control Variables



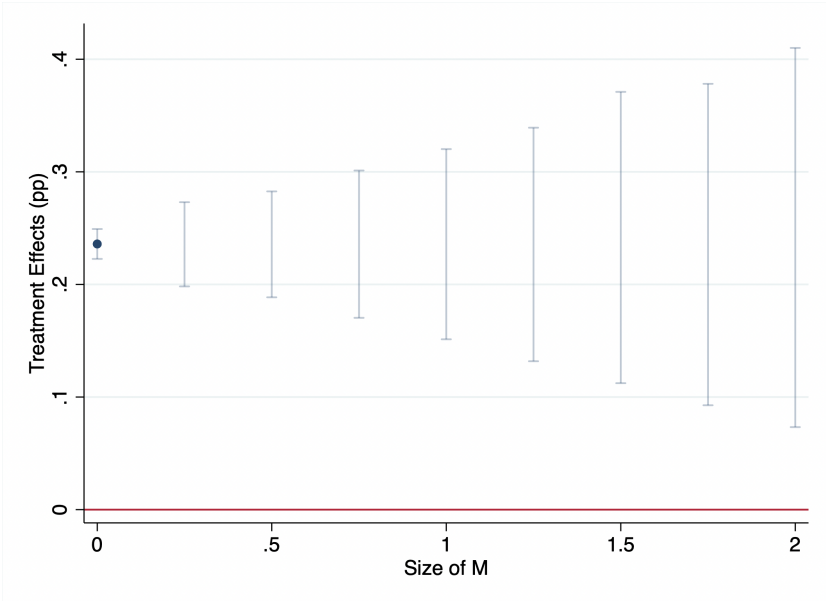
(a) Effects on Offending Initiation



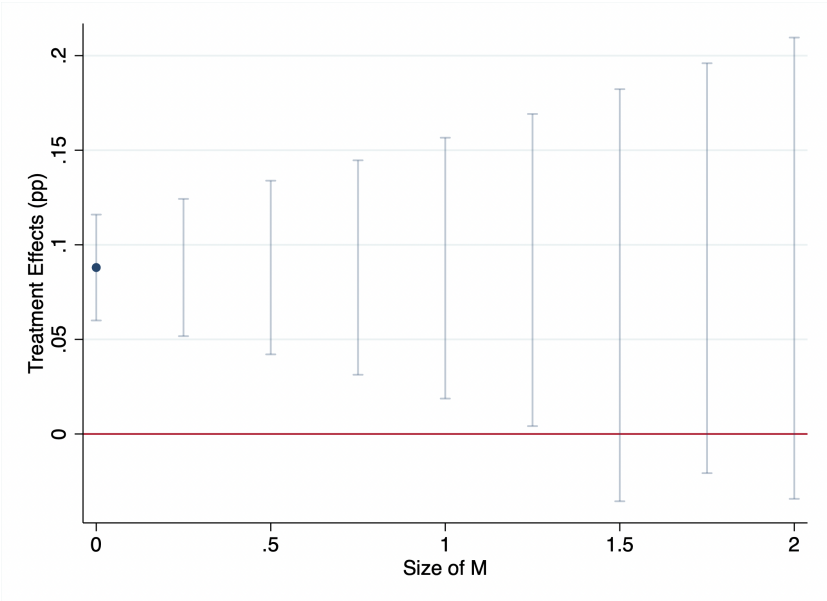
(b) Effects on Number of Crimes

back

Honest DID: Relative Magnitude w/o Controls



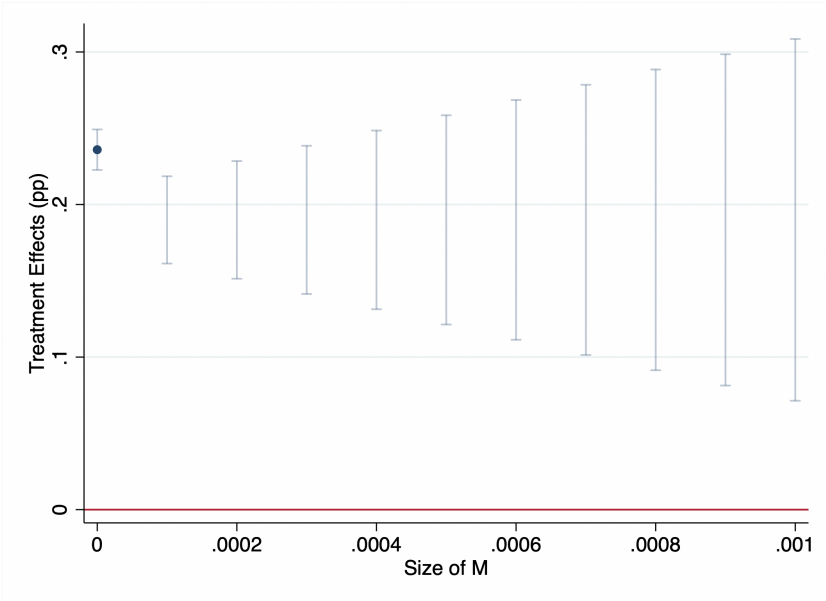
(a) Effects on Offending Initiation



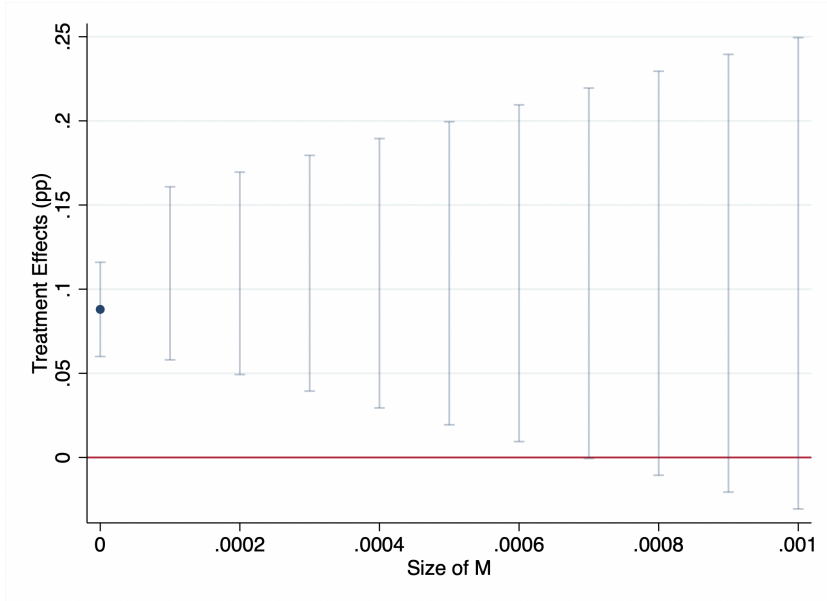
(b) Effects on Number of Crimes

back

Honest DID: Second Differences w/o Controls



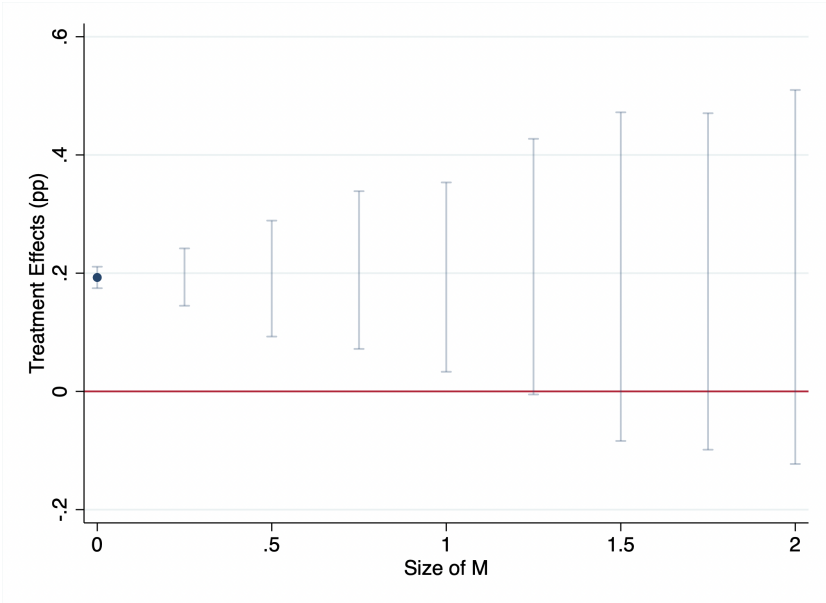
(a) Effects on Offending Initiation



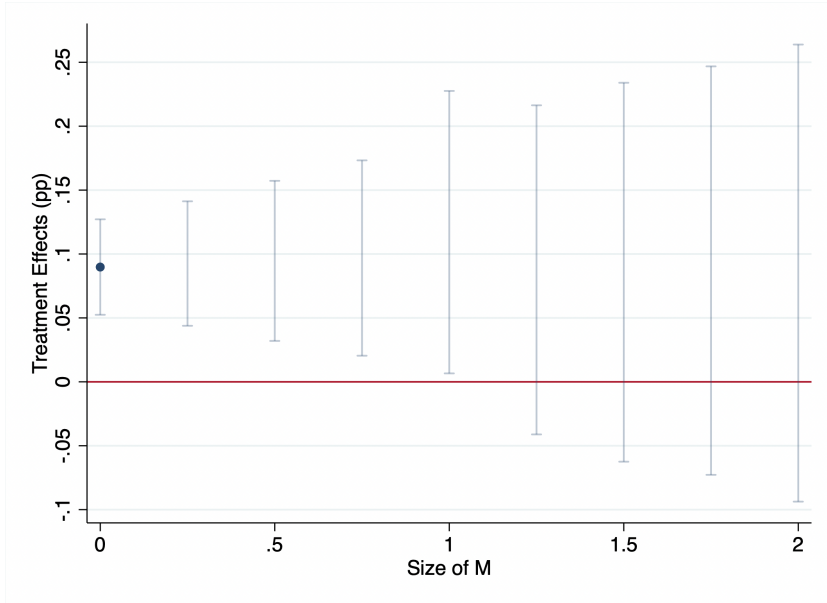
(b) Effects on Number of Crimes

back

Honest DID: Relative Magnitude



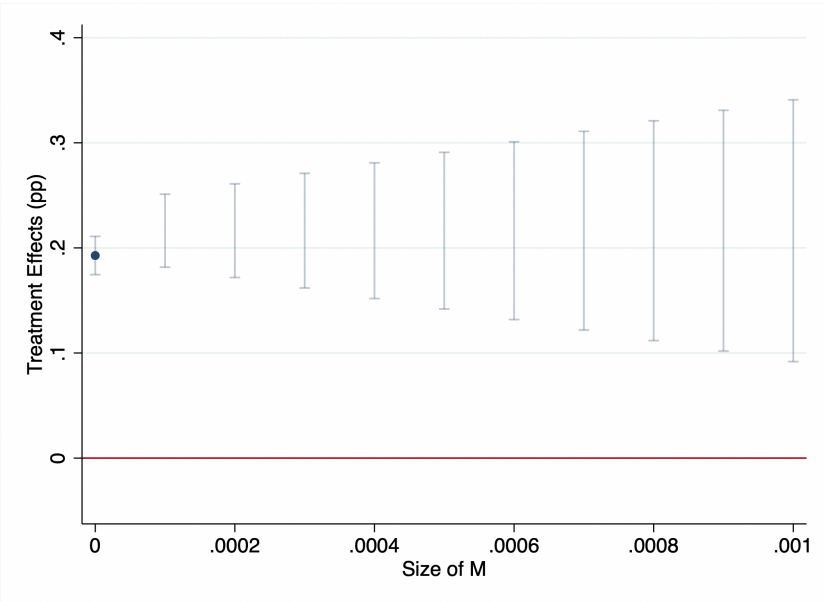
(a) Effects on Offending Initiation



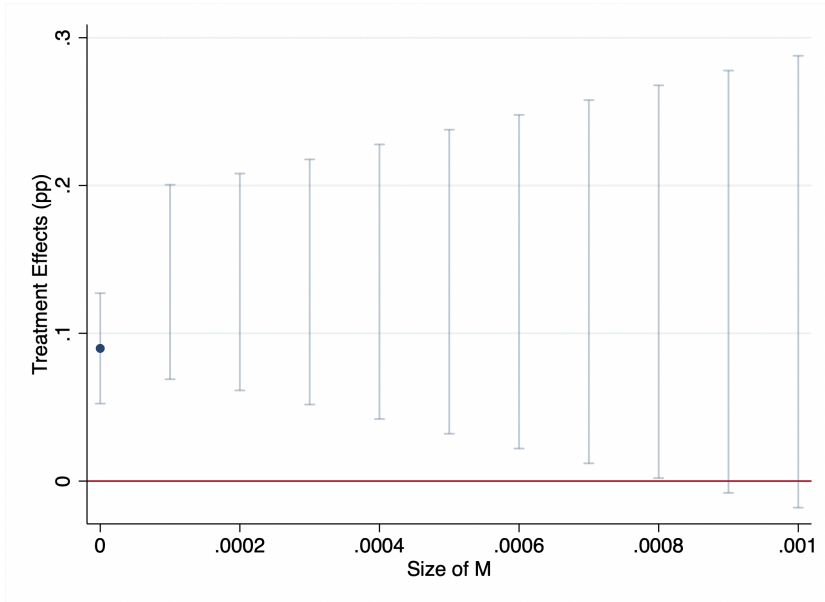
(b) Effects on Number of Crimes

back

Honest DID: Second Differences



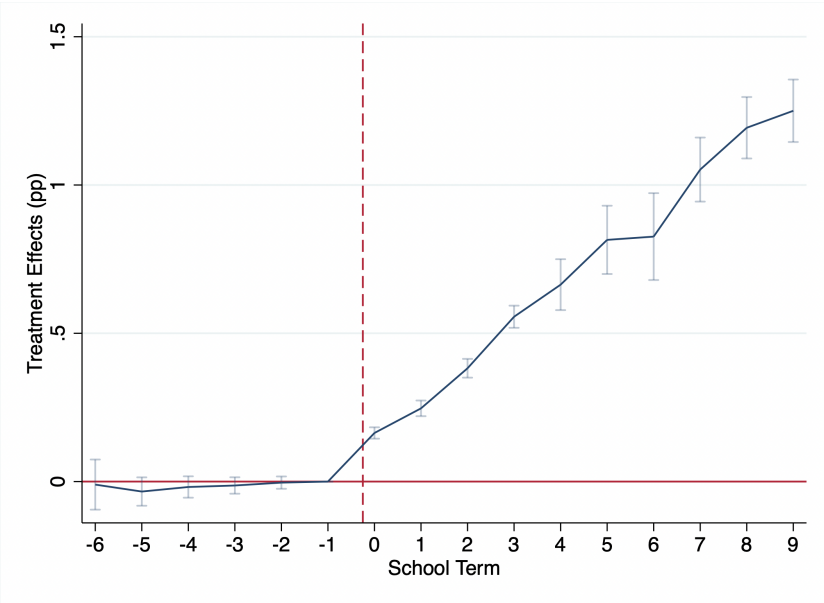
(a) Effects on Offending Initiation



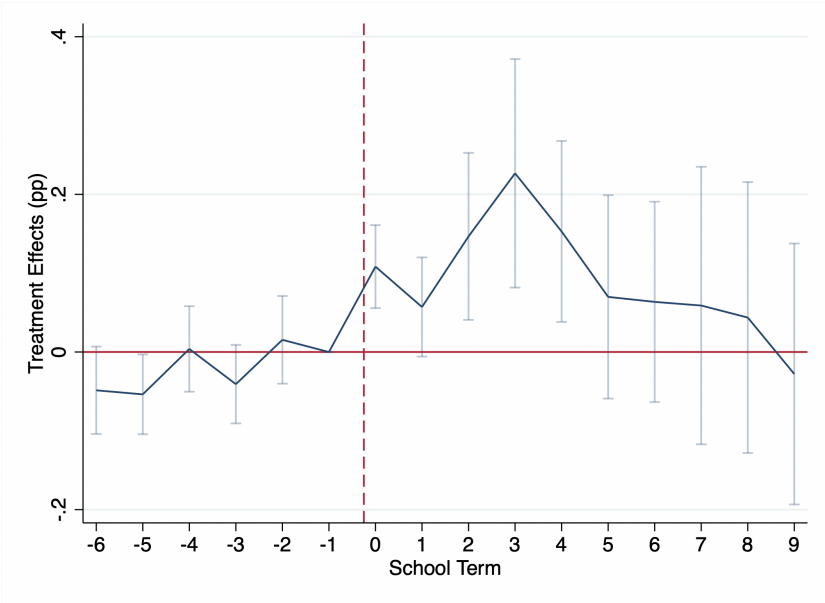
(b) Effects on Number of Crimes

back

Estimation Including Never Treated



(a) Effects on Offending Initiation



(b) Effects on Number of Crimes

back

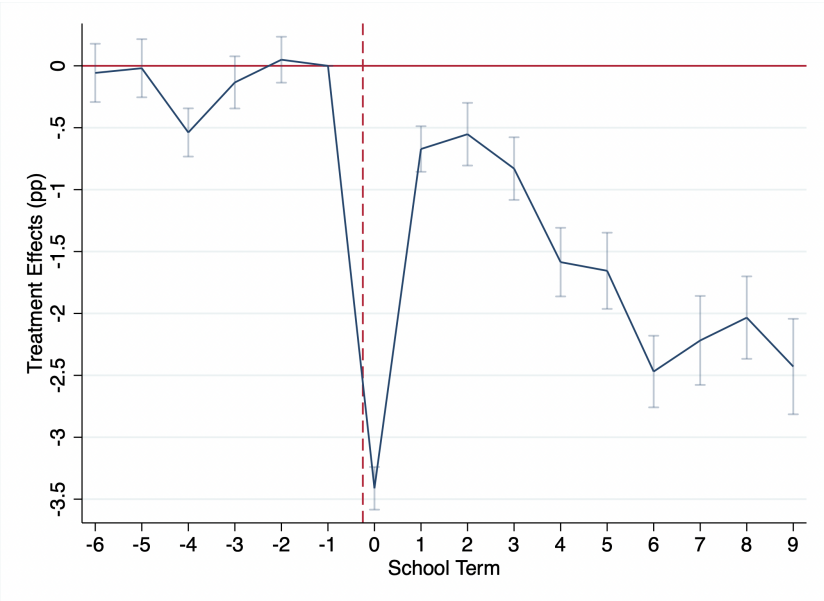
Transfer with a Residence Move

Table 1: Transfer with a Residence Move

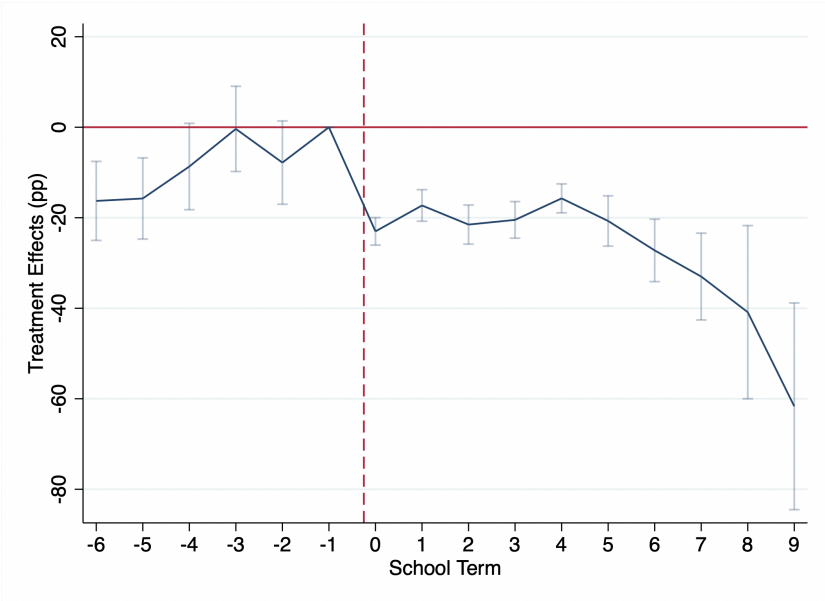
	No Residence Move	Residence Move				
		Any	0-25 miles	25-50 miles	50-100 miles	More than 100 miles
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment Effects at $t = 0$	.182***	.261***	.277***	.346***	.086	.276***
	(.012)	(.021)	(.028)	(.089)	(.055)	(.064)
Pre-Treatment Mean	1.50	1.53	1.53	1.53	1.63	1.58
Relative Effect Size	12.1%	17.0%	18.1%	22.6%	5.26%	17.4%
Sample Size (Panel Count)	10,837,428	4,939,848	3,130,345	731,958	568,499	597,536
Number of Individuals	736,373	335,947	212,850	49,722	38,635	40,797

back

Effects on School Transfers



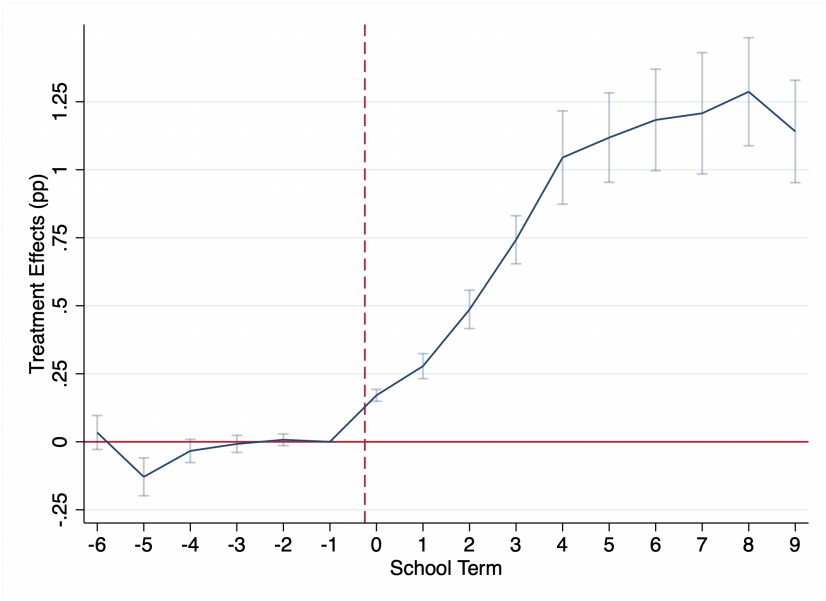
(a) Effects of Receiving an Offender Transfer



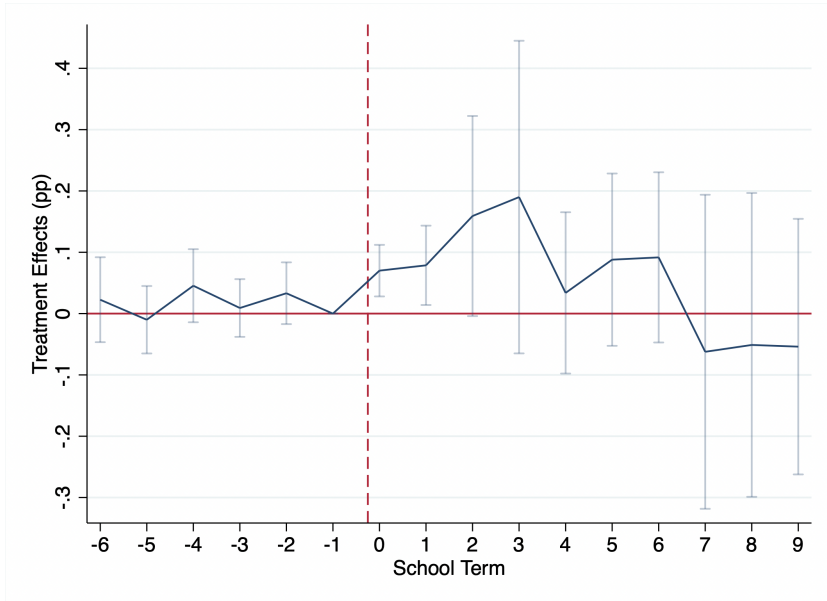
(b) Effects of Receiving a Non-offender Transfer

back

Excluding Those Who Move Back to the Previous School



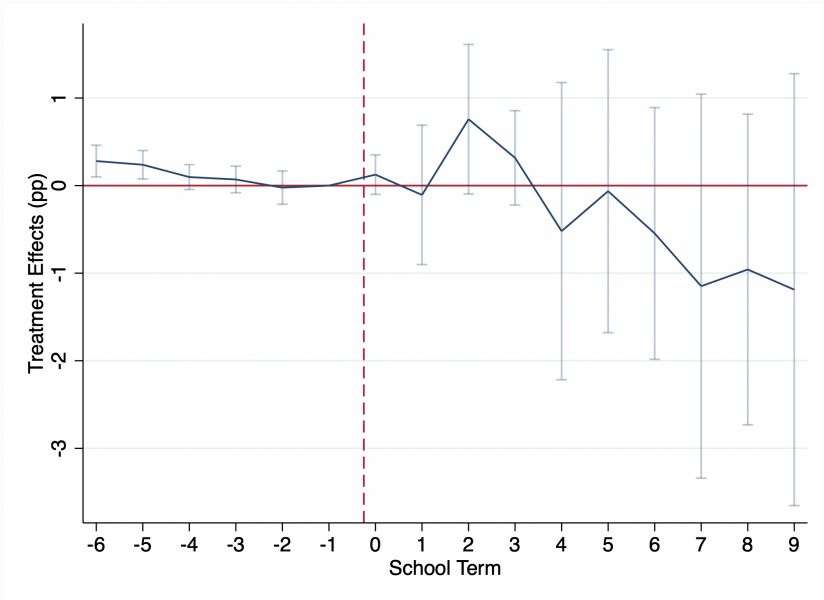
(a) Effects on Offending Initiation



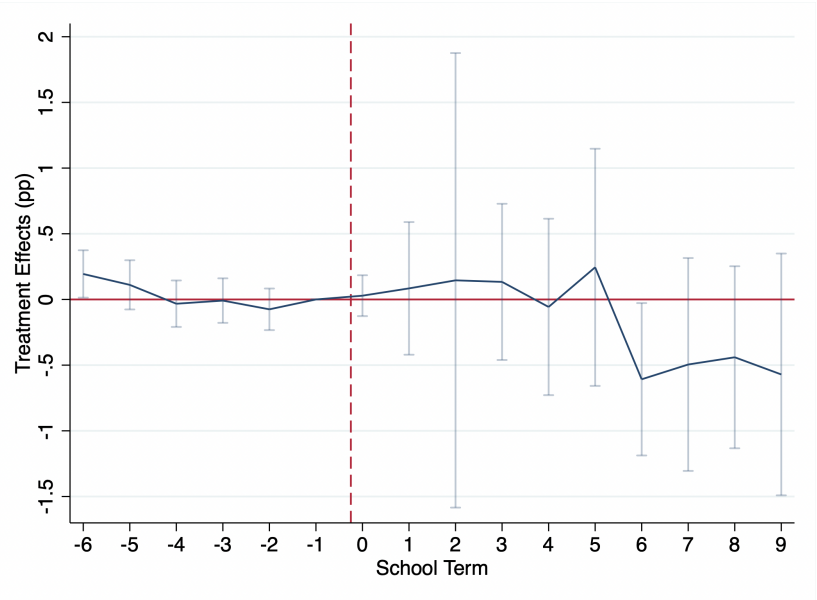
(b) Effects on Number of Crimes

back

Not-yet-offenders' transfers



(a) Effects on Offending Initiation



(b) Effects on Number of Crimes

Back

Effects from Disruptive/Disadvantaged Students

Transfer Students:	Offenders	Non-offenders				
Characteristics of Transfer Students:	Any	FSM	SEN	FSM and SEN	Suspension	FSM, SEN, and Suspension
	(1)	(2)	(3)	(4)	(5)	(6)
Effects at $t = 0$ (x100)	.193***	.075***	.043***	.114***	.106***	.122***
	(.009)	(.008)	(.009)	(.010)	(.008)	(.012)
Control Mean (x100)	1.50	.632	.640	.744	1.22	1.39
Relative Effect Size	12.9%	11.9%	6.79%	15.3%	8.72%	8.74%
Sample Size (Panel Count)	15,458,144	14,409,248	14,286,249	13,819,441	12,551,511	10,406,529
Number of Individuals	1,050,567	979,095	970,886	939,161	852,880	707,334

Back

School vs. Locality

Locality Measure:	School	MSOA			LSOA		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treatment Effects at $t = 0$ (x100)	.193***	.138***	.141***	.070**	.168***	.130***	.050
	(.009)	(.009)	(.027)	(.030)	(.018)	(.023)	(.078)
Control Mean (x100)	1.50	1.51	1.35	1.25	2.16	1.59	1.46
Relative Effect Size	12.9%	9.12%	10.4%	5.60%	7.78%	8.17%	3.42%
School	Same	Any	Same	Different	Any	Same	Different
Locality	Any	Same	Different	Same	Same	Different	Same
Sample Size (Panel Count)	15,458,144	16,617,101	9,883,495	141,222	5,969,939	5,636,557	502,607
Number of Individuals	1,050,567	1,127,226	668,104	738,093	406,544	347,076	31,942

- MSOA: 5,000 to 15,000 residents
- LSOA: 1,000 to 3,000 residents

Back

Crime during School Terms

	During School Term (1)	During Holidays (2)	Weekdays (3)	Weekends (4)
Treatment Effects at $t = 0$	.136***	.090***	.150***	.073***
	(.008)	(.007)	(.009)	(.006)
Pre-Treatment Mean	.997	.665	1.14	.501
Relative Effect Size	13.6%	13.5%	13.1%	14.5%
Sample Size (Panel Count)	15,458,144	15,458,144	15,458,144	15,458,144
Number of Individuals	1,050,567	1,050,567	1,050,567	1,050,567

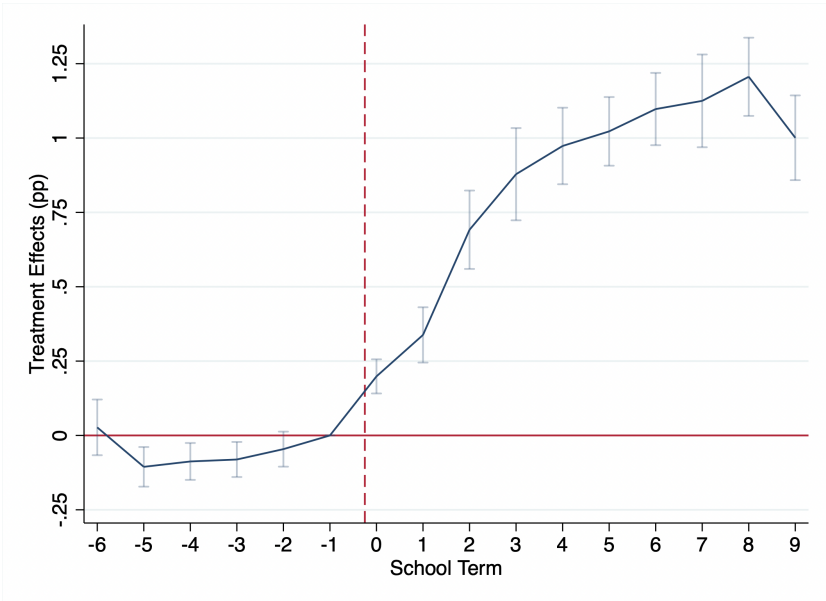
Back

Effects on Stayers

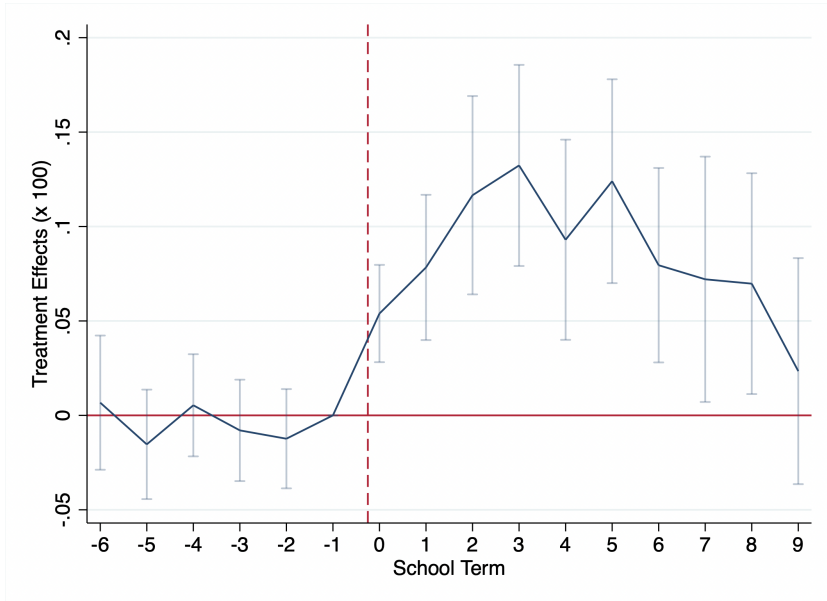
Students Transferring Out:	Offender (1)	Not-yet-offender (2)	Non-offender (3)
Panel A: Offending Initiation			
Treatment effects at $t = 0$	.137***	.117***	.087***
	(.010)	(.041)	(.022)
Pre-Treatment Mean	1.21	.760	.405
Relative Effect Size	11.3%	15.4%	21.5%
Panel B: Number of Crimes			
Treatment effects at $t = 0$	.014	.086***	-.011
	(.021)	(.030)	(.043)
Pre-Treatment Mean	.328	.222	.159
Relative Effect Size	4.32%	38.9%	–7.22%
Sample Size (Panel Count)	12,072,353	15,880,088	14,852,746
Number of Individuals	819,109	1,077,086	1,006,456

▸ back to robustness ▸ back to simulation

Effects on Those in School from which no offenders left



(a) Effects on Offending Initiation



(b) Effects on Number of Crimes

back

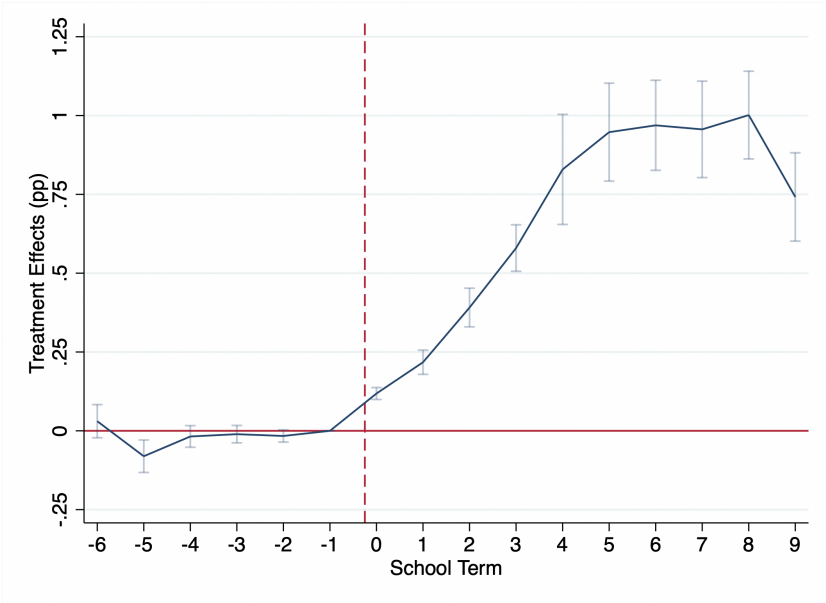
Poisson and Logit Regressions

Table 2: Different Specifications

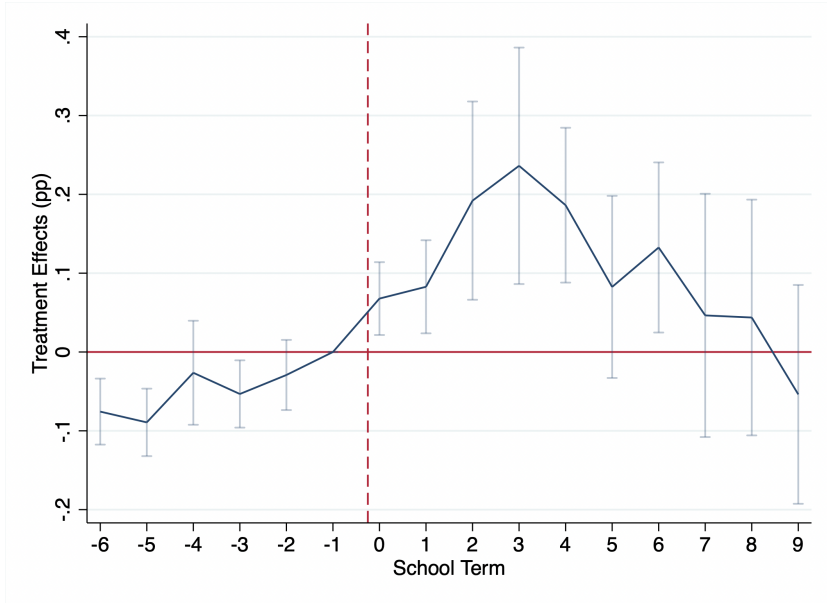
Effects on:	Crime Onset (1)	Any Crime (2)	Number of Crimes (3)
Poisson Regression	.130***	.021***	.030**
	(.010)	(.007)	(.013)
Logit Regression	14.55***	13.28***	-
	(1.38)	(2.41)	
Sample Size	15,262,170	15,262,170	15,262,170

back

Effects Excluding Mass Transfers



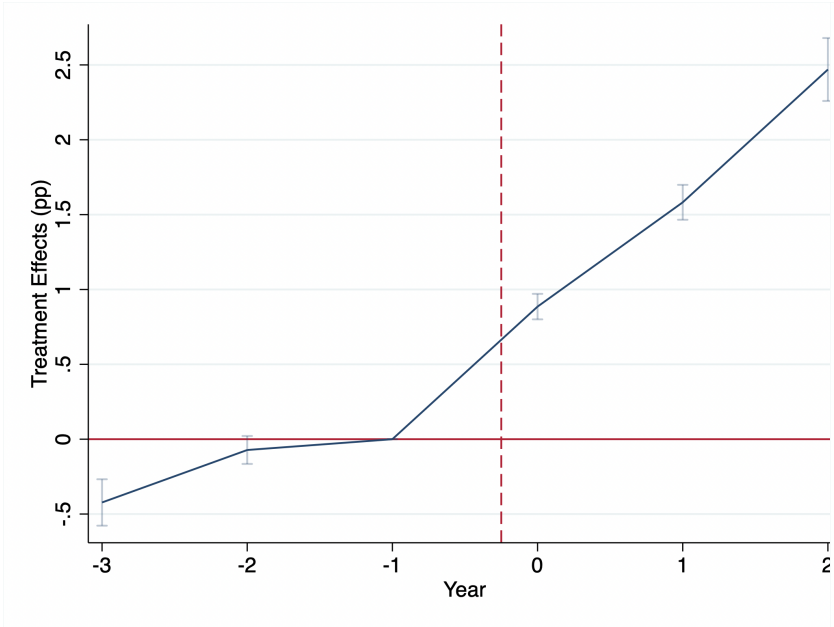
(a) Effects on Offending Initiation



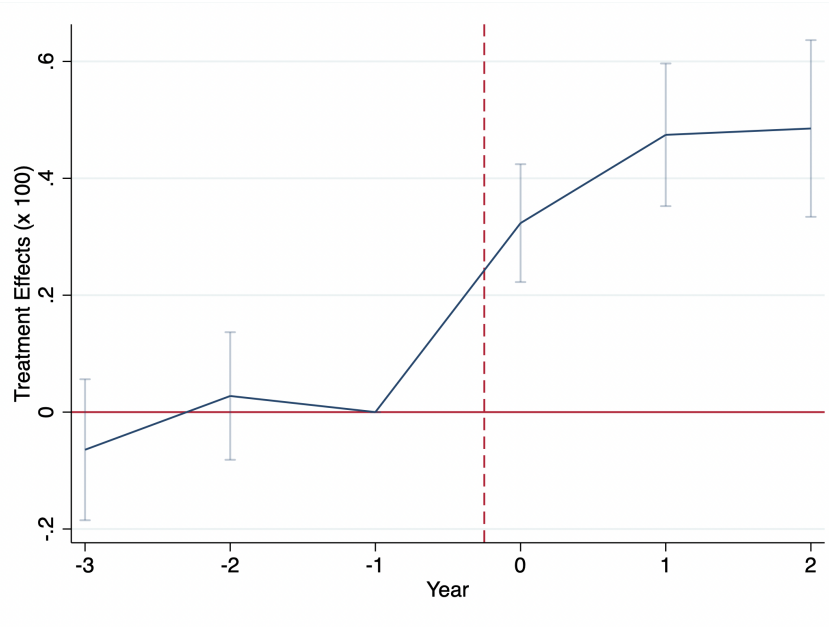
(b) Effects on Number of Crimes

back

Main Results for 1991-94 Cohorts



(a) Crime Onset



(b) Number of Crimes at Each Period

Back

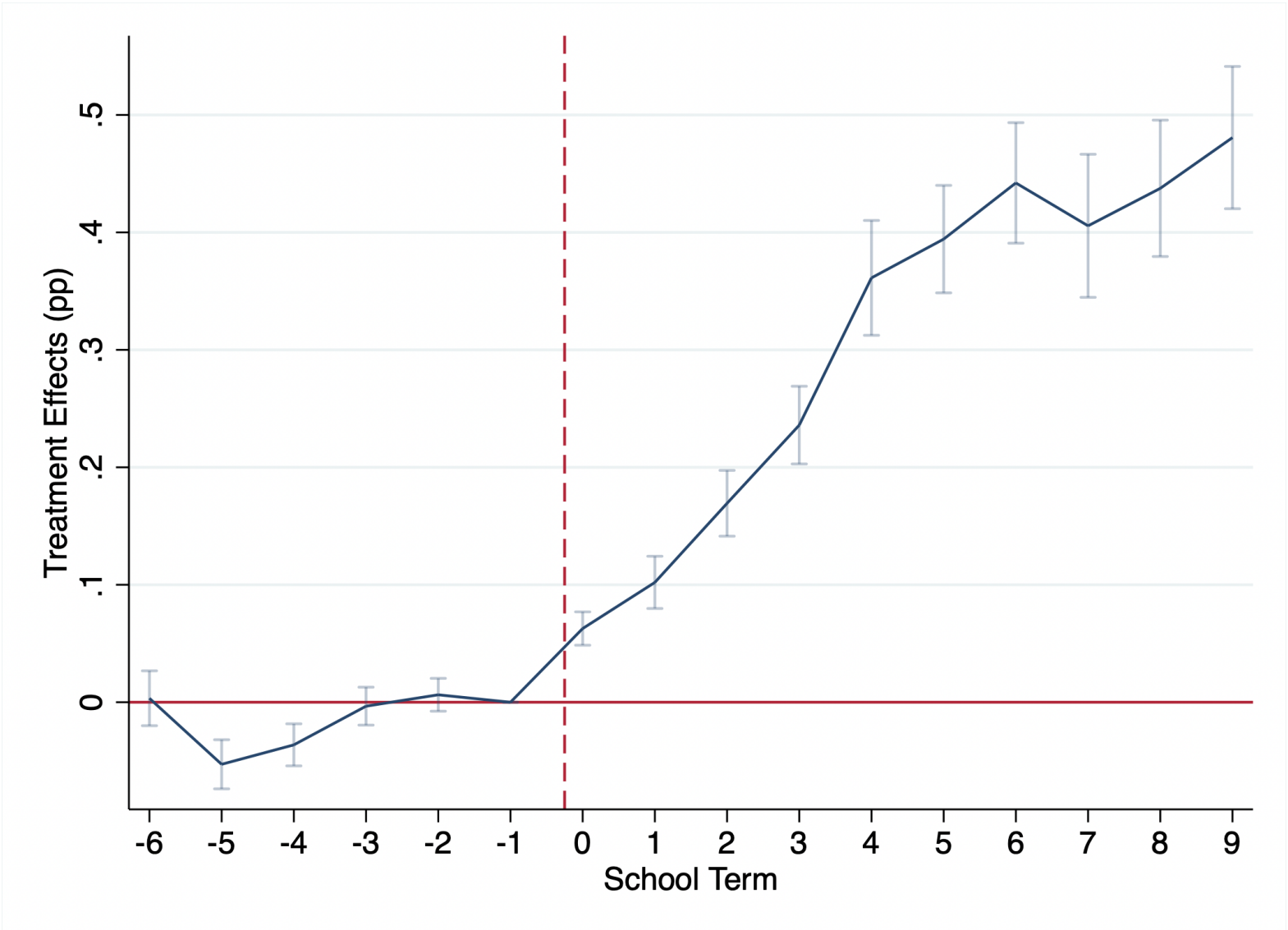
Social Cost

Table 3: Social Costs

	Number of Youth Crimes (1)	Social Cost per Crime (2)	Social Cost of the Crime Type (3)	Increase in the Number Crimes due to Peer Effects (4)	Increase in Soc due to Peer E (5)
All Crimes	61,538	6,456	397,267,948	2,922	20,000,72
Cost of Crime for Each Crime Type					
Violence with Injury	16,233	14,050	228,073,648	355	4,990,29
Violence without Injury	4,125	5,930	24,461,250	940	5,571,98
Rape	426	39,360	16,767,360	78.9	3,103,71
Other Sexual Offence	2,595	6,520	16,919,400	123	799,296
Robbery	2,964	11,320	33,552,480	-42.0	-475,68
Domestic Burglary	4,025	5,930	23,868,250	513	3,041,08
Theft of Vehicle	711	10,290	7,316,190	247	2,542,53
Theft from Vehicle	653	870	568,110	-119	-103,15
Theft from Person	17,080	1,380	23,570,400	449	618,974
Criminal Damage – Arson	720	8,420	6,062,400	-83.7	-704,63
Criminal Damage – Other	10,612	1,350	14,326,200	444	598,877
Fraud	1,344	1,290	1,733,760	0	0
Commercial Theft	50	970	48,500	18.0	17,430

Back

Unauthorized Absences



Back

More Robustness Checks

- Effects from disruptive/disadvantaged pupils' transfers
- Censored at the second treatment
- Estimation without controls
- Bound considering potential violation of pre-trend assumption

  (Rambachan and Roth, 2023)
- Exclude offenders who move back
- School from which no offenders left
- Poisson and Logit regressions
- Drop mass transfers

back

Educational Achievement at Age 11

Quartile in Educational Achievement:	Highest (1)	2nd Highest (2)	3rd Highest (3)	Lowest (4)
Treatment Effects at $t = 0$	.032***	.085***	.149***	.179***
	(.010)	(.015)	(.020)	(.022)
Relative Effect Size	.573	1.13	1.78	2.81
Ratio	5.66%	7.50%	8.38%	6.36%
Sample Size (Panel Count)	3,007,894	2,998,662	2,991,216	2,980,809
Number of Individuals	203,447	203,447	203,447	203,446

Back

FSM/SEN

	With FSM and SEN (1)	With FSM Only (2)	With SEN Only (3)	Neither FSM nor SEN (4)
Treatment Effects at $t = 0$	.570***	.233***	.116***	.072***
	(.041)	(.029)	(.019)	(.008)
Pre-Treatment Mean	4.17	2.04	1.37	.578
Relative Effect Size	13.7%	11.4%	8.47%	12.4%
Sample Size (Panel Count)	2,357,011	2,199,327	3,277,133	7,624,673
Number of Individuals	161,381	150,051	222,544	516,591

Back

Deprivation

Quartile in Deprivation Index:	Least Deprived (1)	2nd Least Deprived (2)	3rd Least Deprived (3)	Most Deprived (4)
All Crimes	.073***	.119***	.209***	.309***
	(.016)	(.017)	(.023)	(.021)
Pre-Treatment Mean	.784	1.10	1.59	2.22
Relative Effect Size	9.35%	10.8%	13.1%	13.9%
Theft	.014	.048***	.082***	.131***
	(.010)	(.010)	(.015)	(.014)
Pre-Treatment Mean	.326	.462	.655	.934
Relative Effect Size	4.31%	10.4%	12.6%	14.0%
Fraud	.002	.001	.005	.008*
	(.002)	(.003)	(.006)	(.004)
Pre-Treatment Mean	.011	.026	.028	.049
Relative Effect Size	21.1%	1.96%	18.2%	15.7%
Sample Size (Panel Count)	3,299,643	3,503,179	3,970,134	4,685,188
Number of Individuals	223,594	237,727	270,064	319,182

▸ All types ▸ Back

Gender

Population:	All (1)	Boys (2)	Girls (3)
Panel A: Effects of Any Offenders' Transfers			
Treatment Effects at $t = 0$	.186***	.242***	.119***
	(.009)	(.014)	(.011)
Pre-Treatment Mean	1.50	1.98	.976
Relative Effect Size	12.5%	12.2%	12.2%
Sample Size (Panel Count)	15,577,130	8,078,484	7,498,646
Number of Individuals	1,058,610	548,665	509,945
Panel B: Effects of Transfers of Offenders of the Same Gender			
Treatment Effects at $t = 0$	.210***	.256***	.122***
	(.013)	(.017)	(.020)
Pre-Treatment Mean	1.62	1.92	1.03
Relative Effect Size	13.0%	13.3%	11.8%
Sample Size (Panel Count)	8,519,492	5,587,964	2,931,528
Number of Individuals	578,696	379,368	199,328
Panel C: Effects of Transfers of Offenders of the Opposite Gender			
Treatment Effects at $t = 0$	.147***	-.814***	.116***
	(.055)	(.151)	(.013)
Pre-Treatment Mean	1.35	2.10	.938
Relative Effect Size	10.9%	–38.8%	12.3%
Sample Size (Panel Count)	7,057,638	2,490,520	4,567,118
Number of Individuals	479,914	169,297	310,617

Back

Language

Heterogeneity in First Language

Population:	Native English Speakers (1)	Non-native English Speakers (2)
Panel A: Effects of Transfers of Offenders with the Same First Language Status		
Treatment Effects at $t = 0$	.503***	.478***
	(.013)	(.108)
Pre-Treatment Mean	1.34	.540
Relative Effect Size	37.5%	88.6%
Sample Size (Panel Count)	14,601,647	416,685
Number of Individuals	834,068	21,763
Panel B: Effects of Transfers of Offenders with a Different First Language Status		
Treatment Effects at $t = 0$	-.304	.007
	(.226)	(.028)
Pre-Treatment Mean	2.01	.883
Relative Effect Size	–15.1%	.804%
Sample Size (Panel Count)	948,034	1,315,063
Number of Individuals	44,262	76,774

Back

Ethnicity

Heterogeneity in Ethnicity

Population:	White British (1)	Non White British (2)
Panel A: Effects of Transfers of Offenders of the Same Ethnicity		
Treatment Effects at $t = 0$	-.290**	.048
	(.129)	(.137)
Pre-Treatment Mean	1.54	1.11
Relative Effect Size	–18.8%	4.35%
Sample Size (Panel Count)	1,856,895	1,346,979
Number of Individuals	126,035	91,035
Panel B: Effects of Transfers of Offenders of a different Ethnicity		
Treatment Effects at $t = 0$	.255***	.176***
	(.037)	(.013)
Pre-Treatment Mean	1.27	1.59
Relative Effect Size	20.1%	11.1%
Sample Size (Panel Count)	1,574,950	10,895,631
Number of Individuals	107,049	741,223

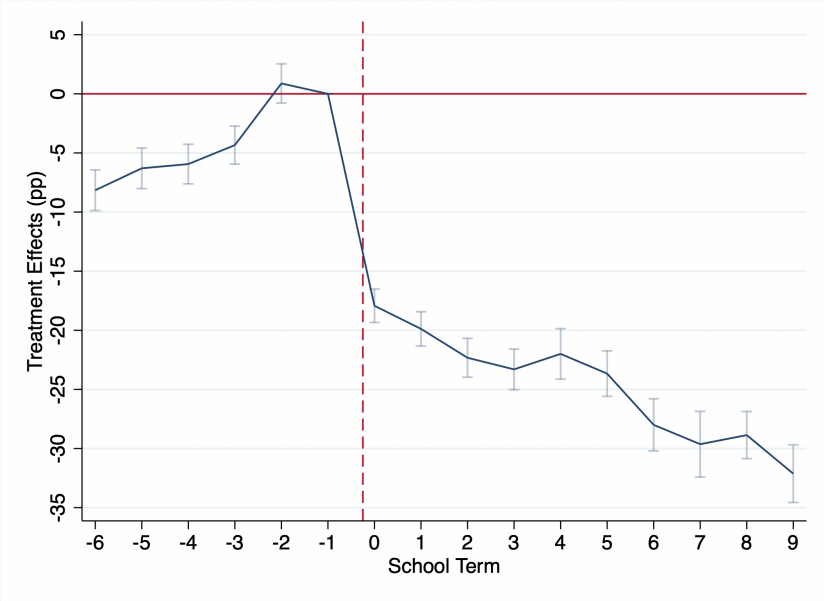
Back

Offender's Experience

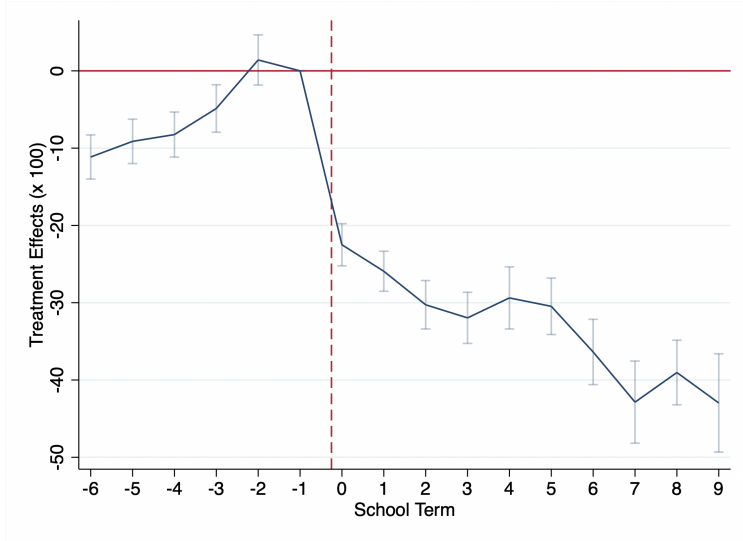
Quartile in offender density:	One Crime (1)	Two to Five Crimes (2)	More than Five Crimes (3)
Panel A: Effects on becoming an offender by each term			
Treatment Effects at $t = 0$	.198***	.275***	.539***
	(.012)	(.024)	(.165)
Pre-Treatment Mean	1.46	1.65	2.10
Relative Effect Size	13.5%	16.7%	25.7%
Panel B: Effects on the number of crimes committed in each term			
Treatment Effects at $t = 0$	.085***	.154***	.474*
	(.023)	(.041)	(.259)
Pre-Treatment Mean	.365	.385	.552
Relative Effect Size	23.4%	39.9%	86.0%
Sample Size (Panel Count)	11,380,413	4,191,421	341,441
Number of Individuals	773,254	285,159	23,257

Back

Behaviors of Transferring Offenders



(a) Effects on Any Offending

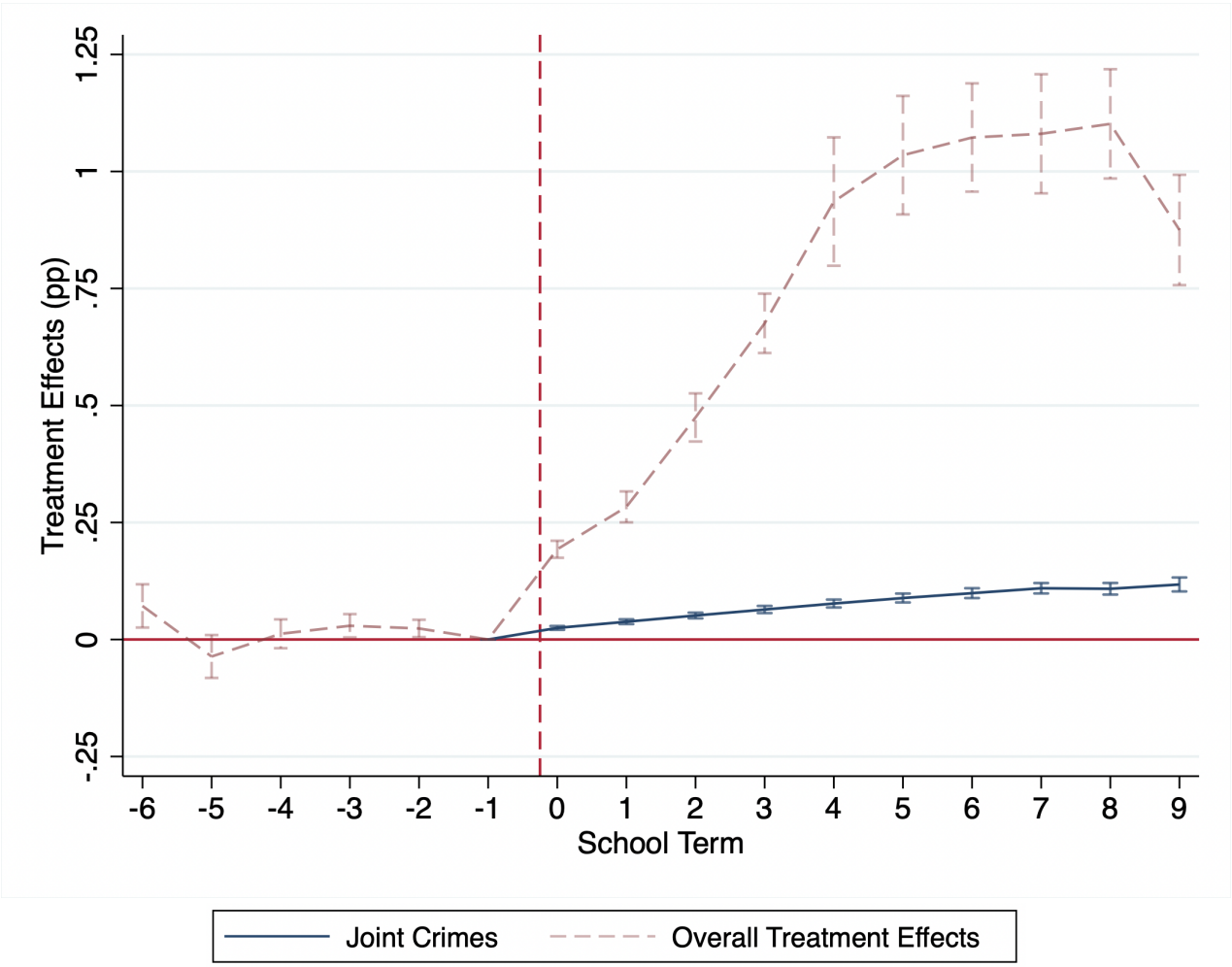


(b) Effects on Number of Crimes

Back

Joint Crimes

- Peers' offense initiation by committing crime on the same day as transfer offenders after they moved



Crime Types, Details

Crime Types:	All Crimes		Violent Crimes		Property Crimes		Drug Crimes		Other Crimes	
	Same (1)	Different (2)	Same (3)	Different (4)	Same (5)	Different (6)	Same (7)	Different (8)	Same (9)	Different (10)
Treatment Effects at $t = 0$	.089***	.082***	.068***	.104***	.106***	.082***	.019	.179**	-.027	.054
	(.007)	(.008)	(.011)	(.014)	(.010)	(.009)	(.034)	(.072)	(.018)	(.046)
Pre-Treatment Mean	.714	.806	.661	1.30	1.08	.714	.121	2.21	.155	1.96
Relative Effect Size	12.4%	10.2%	10.2%	7.98%	9.82%	11.5%	15.8%	8.08%	–17.7%	2.77%
Sample Size (Panel Count)	12,184,487	12,184,487	6,182,281	6,182,281	8,351,676	8,351,676	699,820	699,820	1,628,495	1,628,495
Number of Individuals	827,507	827,507	420,119	420,119	567,629	567,629	47,586	47,586	110,664	110,664

Back

Detail School Types

School Types:	Academy Converter (1)	Academy Sponsor Led (2)	Community School (3)	Foundation School (4)	Voluntary Aided School (5)	Voluntary Controlled School (6)
Panel A: Effects on Crime Onset						
Treatment Effects at $t = 0$	-.027	.104	.139***	-.023	.649*	.467**
	(.035)	(.066)	(.014)	(.301)	(.386)	(.229)
Pre-Treatment Mean	1.12	2.14	1.50	1.61	1.15	1.37
Relative Effect Size	–2.39%	4.86%	9.21%	–1.40%	56.5%	34.1%
Panel B: Effects on Number of Crimes						
Treatment Effects at $t = 0$	-.025	-.122	.049*	-.136	.782***	.247*
	(.088)	(.095)	(.029)	(.303)	(.230)	(.147)
Pre-Treatment Mean	.334	.521	.353	.373	.341	.314
Relative Effect Size	–7.40%	–23.4%	13.9%	–36.3%	229%	78.4%
Sample Size (Panel Count)	2,000,104	989,257	5,710,743	3,747,099	1,272,572	285,724
Number of Individuals	134,819	68,313	386,644	252,803	85,623	19,495

Standard errors reported in parentheses are clustered at the treatment level.

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Early Years

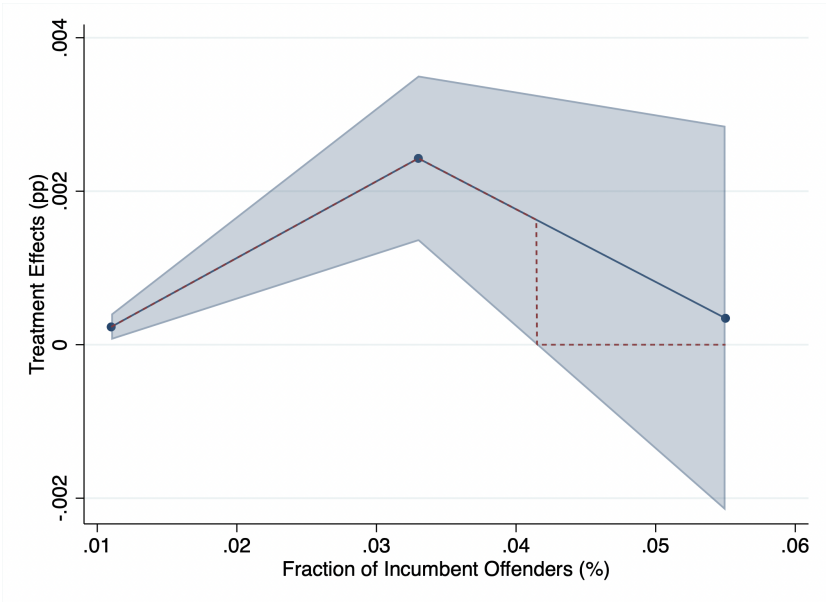
Educational Achievement: Treatment Effects at $t = 0$	High (1)	Low (2)
Overall Test Score	.115**	.129*
	(.049)	(.072)
Pre-Treatment Mean	.675	1.40
Relative Effect Size	17.1%	9.23%
Personal, social and emotional development	.054	.191***
	(.068)	(.069)
Pre-Treatment Mean	.703	1.38
Relative Effect Size	7.66%	13.9%
Communication, language and literacy	.117**	.138*
	(.048)	(.074)
Pre-Treatment Mean	.692	1.39
Relative Effect Size	16.9%	9.93%
Mathematical development	.088	.155**
	(.055)	(.072)
Pre-Treatment Mean	.764	1.32
Relative Effect Size	11.5%	11.7%
Knowledge and understanding of the world	.168***	.079
	(.058)	(.063)
Pre-Treatment Mean	.889	1.18
Relative Effect Size	18.9%	6.72%
Physical development	.112*	.151**
	(.066)	(.061)
Pre-Treatment Mean	.914	1.16
Relative Effect Size	12.2%	13.0%
Creative development	.102*	.158**
	(.059)	(.069)
Pre-Treatment Mean	.863	1.21
Relative Effect Size	11.8%	13.1%
Sample Size (Panel Count)[*1]	273,240	272,122
Number of Individuals[*1]	18,525	18,523

School Types

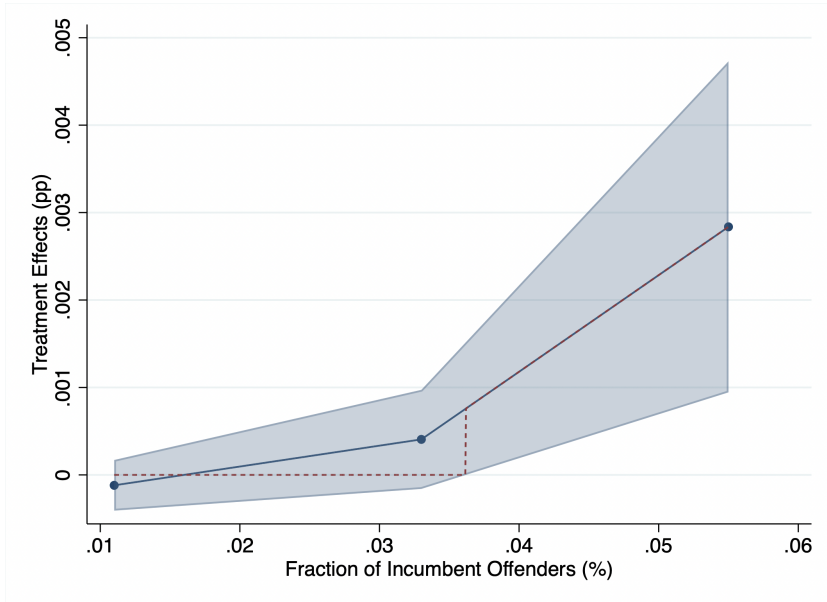
School Types:	Academy Converter (1)	Academy Sponsor Led (2)	Community School (3)	Foundation School (4)	Voluntary Aided School (5)	Voluntary Controlled School (6)
Panel A: Effects on becoming an offender by each term						
Treatment Effects at $t = 0$	-.027	.104	.139***	-.023	.649*	.467**
	(.035)	(.066)	(.014)	(.301)	(.386)	(.229)
Pre-Treatment Mean	1.12	2.14	1.50	1.61	1.15	1.37
Relative Effect Size	–2.39%	4.86%	9.21%	–1.40%	56.5%	34.1%
Panel B: Effects on committing a crime in each term						
Treatment Effects at $t = 0$	-.025	-.122	.049*	-.136	.782***	.247*
	(.088)	(.095)	(.029)	(.303)	(.230)	(.147)
Pre-Treatment Mean	.334	.521	.353	.373	.341	.314
Relative Effect Size	–7.40%	–23.4%	13.9%	–36.3%	229%	78.4%
Sample Size (Panel Count)	2,000,104	989,257	5,710,743	3,747,099	1,272,572	285,724
Number of Individuals	134,819	68,313	386,644	252,803	85,623	19,495

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Simulation: Offenders' Transfers



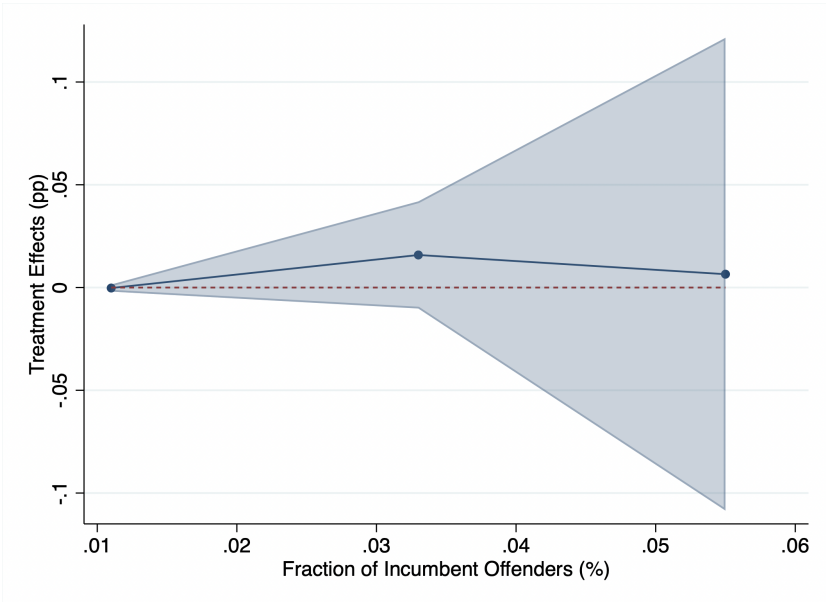
(a) Effects on Offending Initiation



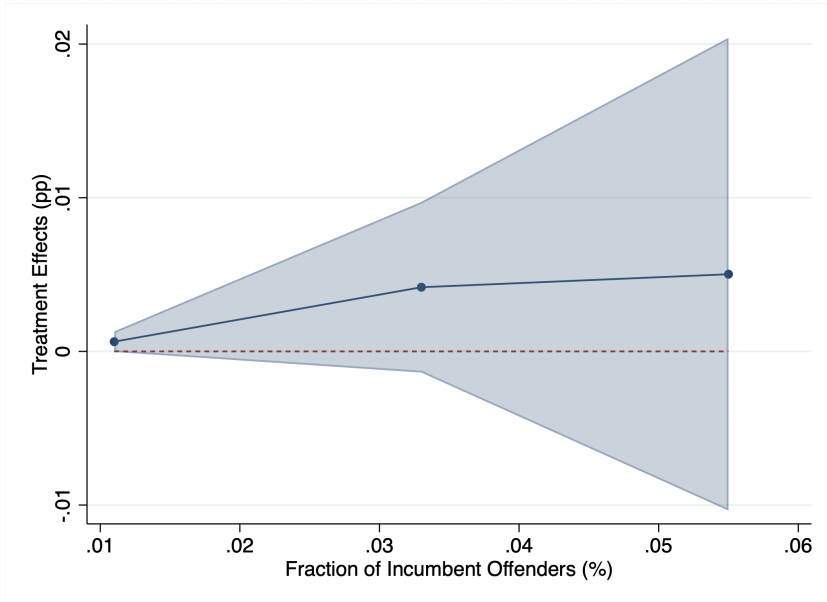
(b) Effects on Number of Crimes

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Simulation: Non-offenders' Transfers



(a) Effects on Offending Initiation



(b) Effects on Number of Crimes

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Simulation

		Swapping Radius (miles)					
	Status quo	1	3	5	10	25	Any
Panel A: Minimizing N of Crimes							
N of offenders swapped	3661	310	1649	2283	3071	3543	3657
Fraction of offenders swapped		8%	45%	62%	84%	97%	100%
N of crimes	379	350	244	193	137	113	91
Δ crimes		-8%	-36%	-49%	-64%	-70%	-76%
N of offenders	887	893	917	932	924	918	826
Fraction of offenders		1%	3%	5%	4%	3%	-7%
Panel B: Minimizing N of Offenders							
N of offenders swapped	3661	309	1579	2224	2961	3470	3653
Δ offenders swapped		8%	43%	61%	81%	95%	100%
N of crimes	379	392	498	609	805	1089	1244
Δ crimes		3%	31%	61%	113%	187%	228%
N of offenders	887	848	686	617	574	603	648
Δ offenders		-4%	-23%	-30%	-35%	-32%	-27%

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Deprivation

Quartile in Deprivation Index:	Least Deprived (1)	2nd Least Deprived (2)	3rd Least Deprived (3)	Most Deprived (4)
Effects on becoming an offender by each term				
All Crimes	.073***	.119***	.209***	.309***
	(.016)	(.017)	(.023)	(.021)
Pre-Treatment Mean	.784	1.10	1.59	2.22
Relative Effect Size	9.35%	10.8%	13.1%	13.9%
Theft	.014	.048***	.082***	.131***
	(.010)	(.010)	(.015)	(.014)
Pre-Treatment Mean	.326	.462	.655	.934
Relative Effect Size	4.31%	10.4%	12.6%	14.0%
Battery	.024***	.040***	.068***	.068***
	(.008)	(.008)	(.013)	(.010)
Pre-Treatment Mean	.172	.233	.357	.460
Relative Effect Size	14.2%	17.3%	19.1%	14.9%
Drug	.013***	.014***	.023***	.019***
	(.005)	(.005)	(.006)	(.005)
Pre-Treatment Mean	.024	.032	.043	.040
Relative Effect Size	53.4%	42.0%	53.1%	48.8%
Violence	.006	.018***	.033***	.062***
	(.006)	(.006)	(.010)	(.009)
Pre-Treatment Mean	.083	.128	.215	.316
Relative Effect Size	7.56%	14.0%	15.2%	19.7%
Fraud	.002	.001	.005	.008*
	(.002)	(.003)	(.006)	(.004)
Pre-Treatment Mean	.011	.026	.028	.049
Relative Effect Size	21.1%	1.96%	18.2%	15.7%
Weapon	.006*	.004	.011**	.009*
	(.003)	(.002)	(.005)	(.005)
Pre-Treatment Mean	.025	.029	.048	.077
Relative Effect Size	22.6%	13.5%	22.7%	12.1%
Arson/Damage	.030***	.025***	.062***	.085***
	(.008)	(.009)	(.013)	(.010)
Pre-Treatment Mean	.185	.295	.412	.604
Relative Effect Size	16.2%	8.42%	15.1%	14.0%
Sexual	-.000	.006**	.011***	.007**
	(.002)	(.003)	(.004)	(.004)
Pre-Treatment Mean	.032	.023	.040	.053
Relative Effect Size	-.783%	27.6%	28.3%	13.5%
Motoring	.000	.003	.006*	.010***
	(.003)	(.003)	(.004)	(.004)
Pre-Treatment Mean	.007	.010	.016	.035
Relative Effect Size	4.45%	30.1%	40.9%	29.9%
Others	.004	.008**	.017***	.012**
	(.003)	(.004)	(.004)	(.006)
Pre-Treatment Mean	.023	.034	.042	.073
Relative Effect Size	19.0%	23.3%	41.1%	16.6%
Sample Size (Panel Count)	3,299,643	3,503,179	3,970,134	4,685,188
Number of Individuals	223,594	237,727	270,064	319,182

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